The Viability of Blockchain Markets under Discrete Clearing and Paid Priority

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ABSTRACT

The current design of blockchains may fail to sustain a functioning market. We model three features that distinguish blockchain markets from traditional exchanges: discrete block-time clearing, paid priority execution against passive liquidity, and mechanical price impact. Only traders with valuations beyond an endogenous participation threshold compete for priority. This selection concentrates trading among aggressive traders, distorts prices, raises volatility, and increases the losses of liquidity suppliers. Surprisingly, lower information costs and greater liquidity demand undermine price discovery by intensifying priority competition and concentrating trades among increasingly aggressive valuations. Block time amplifies these frictions and can cause markets to shut down.

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Introduction

The original vision of blockchains is to provide a decentralized infrastructure to organize economic activity at scale.[1] The ability of this infrastructure to support financial markets is therefore a fundamental test of that ambition. This paper develops a theoretical framework that models the blockchain as the sole venue for price formation, and asks whether it can sustain a functioning market. Our model incorporates three defining features that distinguish blockchain markets from traditional electronic markets. First, clearing occurs periodically, batching many transactions into *blocks*. Second, within each block, transactions are executed in sequence through paid-priority discriminatory pricing. Third, liquidity provision is passive and price impact between transactions is mechanical, set by fixed algorithmic rules.

A blockchain stores and executes transactions on a shared ledger that a network of participants updates through *consensus*. Transactions are recorded into *blocks* built at regular intervals, called *slots*, and the duration of a slot is called *block time*. Block time is necessary to reach consensus. At the end of each slot, a *builder* builds the next block, and transactions in the block are executed sequentially in order of *priority fees* that traders pay to the builder. Each transaction is filled against liquidity that sits passively in a decentralized exchange (DEX).[2] Importantly, the price impact of each transaction is mechanical, fixed by the DEX's code rather than by belief updates of a market maker. Traders engage in *priority gas auctions* (PGAs) to compete for better queue positions in the block with the objective of obtaining better execution prices.

Our main findings are threefold. First, paid priority, passive liquidity supply, and mechanical price impact generate endogenous selection, where only traders with sufficiently high valuations transact. This hinders price discovery by biasing prices toward those valuations. Second, because only high-valuation traders transact, adverse selection borne by liquidity

[1] See Harvey (2016), Cong and He (2019) and Biais, Capponi, Cong, Gaur, and Giesecke (2023).

[2] DEXs operate as smart contracts; see John, Kogan, and Saleh (2023). The contract defines algorithmic rules that govern price and liquidity (see Barbon and Ranaldo (2021); Aoyagi and Ito (2021)). In 2025, DEXs supported approximately $420 billion in monthly trading volumes.

suppliers increases so liquidity decreases in equilibrium. Paradoxically, these negative effects intensify, rather than ease, when information is cheaper to acquire, which reverses the comparative statics of traditional markets. Third, block time, though necessary to secure the blockchain, amplifies these effects and can cause the market to shut down.

We model the strategic interaction between liquidity suppliers and liquidity takers for a risky asset on a blockchain in three stages. In stage one, traders decide whether to acquire information at a cost and enter the blockchain. In stage two, a liquidity supplier sets the depth of liquidity in a DEX by balancing revenue from price-sensitive noise demand against expected losses to informed traders (see Malinova and Park (2024)). In stage three, informed traders compete in a PGA and noise traders submit transactions. The valuations of informed traders combine the common value of the asset, which they all observe, with private independent trading motives. Block time is long,[3] so price innovations over a block can be substantial, and informed traders in our model are either all buyers or all sellers. The remainder of the introduction considers buyers; the case with sellers is symmetric.

Traders compete for priority in the block to avoid worse execution prices resulting from the mechanical price impact of preceding transactions in the block. We propose and solve a special form of multi-prize auction for the PGA in which each trader's payoff depends on the rank of their bid relative to that of others. Specifically, the prize from bidding a priority fee is the expected price impact that a trader avoids by improving queue position. In the perfect Bayesian equilibrium of the PGA, informed traders determine both their priority fees and their trading volumes. To determine priority fees, they balance profits from better execution prices against the fee paid to the builder. To determine volumes, they balance increased profits from larger positions against execution costs that increase with transaction size.

Our first finding is that discriminatory pricing due to paid priority, passive liquidity provision, and mechanical price impact together have adverse effects on markets. Specifically, we show that there exists a valuation level (cutoff) below which traders choose not to partic-

[3]Blockchains that host decentralized markets, such as Ethereum, operate with slots of twelve seconds or longer to build blocks securely.

ipate in the PGA. The intuition is as follows. Low valuations of the asset lead to low desired volumes, low volumes lead to weaker incentives to compete in the PGA and to low priority fees, and low priority fees lead to worse queue positions within the block. A low-valuation trader therefore anticipates bearing the price impact of the higher-valuation traders ahead in the block. For traders with sufficiently low valuations, this price impact makes the expected wealth from trading negative, so they do not participate. Hence only informed traders with high enough valuations transact on the blockchain.

Thus, blockchain trading concentrates among aggressive, high-valuation traders. Our second finding shows that this selection impairs liquidity. This is because only high-valuation traders trade, so prices are biased toward their high valuations. The liquidity supplier anticipates the resulting high adverse selection costs and supplies less liquidity. Competition intensifies these effects. As more traders compete, individual trading costs due to paid priority and price impact increase, so higher valuations are required for profitable trading. Therefore, the participation cutoff rises, trading concentrates in an even thinner tail of high-valuation traders, the price is more biased, and liquidity decreases further.

Paradoxically, we show that a lower cost of information or a larger mass of noise demand impairs price formation due to the endogenous selection mechanism in blockchains. Specifically, a lower cost of information raises the ex-ante equilibrium number of informed traders who pay that cost to enter the blockchain and compete in the PGA. The added competition raises the participation cutoff ex post, so prices become more biased.

Our findings contrast with outcomes in traditional electronic markets, where price discovery is continuous and trading unfolds over many short clearing rounds. There, greater competition among informed traders accelerates the incorporation of information, and market depth typically improves over time; see Holden and Subrahmanyam (1992) and Foster and Viswanathan (1996). Blockchain markets differ fundamentally. Each block clears only once and aggregates trading activity that would otherwise unfold over many successive rounds in a traditional electronic market. Moreover, transactions are ordered not by time priority but

by paid priority, with mechanical price impact between them. Due to paid priority, only high-valuation traders are willing to trade, and due to the discrete nature of blocks, the liquidity supplier must absorb the adverse selection risk from the aggregate informed flow in a single decision, rather than adjusting liquidity and prices across successive rounds.

Our third finding is that block time creates a tradeoff between security and efficiency. Longer block times increase the dispersion of private valuations and intensify adverse selection, while simultaneously reducing the effective mass of price-sensitive noise demand. As a result, with many informed traders, liquidity decreases and there is a critical block time beyond which the market collapses.

Our model implies several stylized facts that we take to the data. First, using historical data from the DEX Uniswap v3 on Ethereum, we study assets whose price forms on the blockchain.[4] For these assets, the sequence of transactions within a block reflects on-chain price discovery, and we find that it follows the pattern predicted by the model. First, transactions at the start of the block are predominantly in the same direction. Second, the volume of transactions decreases with queue position. Third, the expected net trading volume is close to zero at the end of the block. This supports the equilibrium structure of a block that our model predicts. Aggressive traders with high valuations obtain the earliest queue positions, followed by traders with lower valuations who trade in the same direction, and then by noise traders who trade in both directions.

Second, we study assets that are traded significantly both on the blockchain and on a centralized venue, so price formation occurs on both markets. For these assets, we find that the price change over a single block is significantly more likely to be extreme on the blockchain than on the centralized venue, so prices are more volatile on blockchain markets. We also find that per-transaction volumes on the blockchain are substantially higher. This supports the model's prediction that competition and priority ordering among informed traders induce selection of high-valuation traders that bias prices toward extreme values.

[4]That is, assets that are not traded on a major centralized electronic exchange.

Literature review. Our paper is closest to two recent contributions on price formation and liquidity provision on blockchains. Malinova and Park (2024) study the design of liquidity supply in a DEX, in a setting without periodic clearing or priority-fee ordering of transactions within a block. We adopt their zero-profit characterization of liquidity supply for our stage two, and our contribution is to embed it in a model of the auction for block priority, which is what generates the paper's central selection mechanism. He, Yang, and Zhou (2025) study priority fees paid by traders who arbitrage the blockchain against a competing venue. In contrast, we ask if the blockchain can serve as the primary venue for price formation, absent an arbitrage relationship to an external market. Azar, Casillas, and Farboodi (2024a,b) study how informational advantages among blockchain intermediaries sustain rents and centralization; we instead focus on the resulting effects on price formation and liquidity supply. Beyond these three papers, our contribution relates to four further strands of the literature.

The tradeoff faced by informed traders in PGAs is analogous to that faced by political contestants or researchers in R&D races; see Lazear and Rosen (1981); Hillman and Riley (1989); Barut and Kovenock (1998). In such contests, the cost of the effort exerted by contestants corresponds to the priority fee paid by traders, and the payoff linked to achieving a particular rank in the race corresponds to the prize in the PGA; see also Klose and Kovenock (2015). Multi-prize contests are typically modeled as independent private cost contests, where the cost of effort is private but the prize vector is known ex ante; see Moldovanu and Sela (2001). In contrast, in the PGA, the *prize* from attaining a given position is not fixed but depends on the private valuations and trading volumes of competitors. In the PGA, the value of a given position corresponds to the price impact that a trader avoids by securing that position within the block.

Our model is related to the literature that studies the properties of price and liquidity. Our model is information based. However, it shares features of the multi-stage inventory-based models in Biais (1993); De Frutos and Manzano (2002). We adapt these models to

account for information dissemination, DEX algorithmic pricing, and the blockchain protocol whereby traders compete for queue position instead of speed (O'Hara (1998)). Our model is also related to the theory on the costs of information processing such as that in Grossman and Stiglitz (1980), Verrecchia (1982), Kyle (1989), and Baldauf and Mollner (2020).

Some studies examine periodic uniform price auctions for market clearing (Madhavan (1992); Budish, Cramton, and Shim (2015)), and they show that these auctions enhance price efficiency but impose higher information costs on traders. Instead, our model considers discriminatory auctions in which liquidity suppliers do not participate. Several studies assume an information monopolist (Kyle (1985)). In blockchains, monopolistic information would drive priority fees to zero in equilibrium, which contradicts empirical evidence showing that priority fees increase with the informational content of transactions; see Capponi, Jia, and Yu (2024). We therefore consider competition among multiple informed traders. Models with competition, such as Holden and Subrahmanyam (1992) and Foster and Viswanathan (1996), lead to revealing equilibria where prices reflect private information. In our model, traders submit fully revealing priority fees, however, blockchains operate with periodic clearing and the revealing equilibrium is delayed until the end of block time.

Trading in blockchains resembles trading in preopening sessions of traditional electronic exchanges, where markets clear only at the end of a short period. Unlike preopening sessions, where orders execute at a uniform clearing price that maximizes volume, blockchain transactions are priced by priority fees, and volume is not maximized. Moreover, traders in preopening sessions observe partial information about the market, such as liquidity at the best prices or a virtual clearing price; see Van Bommel and Hoffmann (2011).[5]

The microstructure of DEXs has been studied extensively. The economic tradeoffs for liquidity supply are similar to those for market makers in limit order books and dealer markets

[5]Biais, Hillion, and Spatt (1999) propose several hypotheses regarding the informativeness of preopening prices; our empirical results in Appendix A show that information revelation is delayed in public memory pools, which supports their *pure-noise hypothesis*, under which trading activity before clearing contains no new information. Similarly, Medrano and Vives (2001) find that trading volumes and information revelation accelerate toward the end of preopening sessions.

(Ho and Stoll (1983); Glosten and Milgrom (1985)), and are characterized for DEXs in Malinova and Park (2024); see also Lehar and Parlour (2025), Hasbrouck, Rivera, and Saleh (2026), Milionis, Moallemi, Roughgarden, and Zhang (2022), and Park (2023).[6] DEXs can be viewed more generally as parameterized price schedules (Glosten (1994); Biais, Martimort, and Rochet (2000)), so the conclusions of our model extend to other market infrastructures that a blockchain might run.[7]

The role of DEXs is likely to grow with the development of tokenization of real-world assets (Heines, Dick, Pohle, and Jung (2021); Cartea, Drissi, and Saleh (2026)), central bank digital currencies (Fung and Halaburda (2016); Auer, Frost, Gambacorta, Monnet, Rice, and Shin (2022)), and cross-border settlement (Harvey (2021); Hub (2023); Cardozo, Fernández, Jiang, and Rojas (2024)).

The remainder of this paper proceeds as follows. In Section I, we present notation, assumptions, and structure of the model. In Section II, we study the priority fee bidding strategies and trading volumes of informed traders. In Section III, we analyze the liquidity supply. In Section IV, we study incentives to acquire information at a cost. In Section V, we study the economic effects of block time. In Appendix A, we describe the features of blockchains that underpin our model. In Appendix B, we discuss extensions of our model. Finally, Appendix C collects the proofs.

I. Model

Blockchains execute and store transactions without a central authority. Instead, a network of participants maintains a shared ledger and appends new transactions to it through *consensus*. Consensus proceeds in rounds, called slots. Throughout each slot, traders submit

[6] See also Angeris, Evans, and Chitra (2021); Capponi, Iyengar, Sethuraman, et al. (2023); Drissi, Wu, and Jaimungal (2025) who show that DEXs cause losses to liquidity suppliers. Klein, Kozhan, Viswanath-Natraj, and Wang (2023) study the role of informed liquidity supply in price discovery. Hasbrouck, Rivera, and Saleh (2023) study concentrated liquidity DEXs. Cartea, Drissi, Sánchez-Betancourt, Siska, and Szpruch (2024b) propose AMM designs to mitigate adverse selection.

[7] See Capponi et al. (2024) for an empirical study of the influence of gas fees on the informativeness of trades.

transactions. At the end of each slot, a block is created, and only then are the submitted transactions executed. When the block is created, transactions are executed sequentially, according to *priority fees* paid by traders who compete for priority. Priority fees are revenue for consensus participants who secure the blockchain; see Appendix A for more details.

Transactions are executed against liquidity that sits in a *decentralized exchange* (DEX) running on the blockchain.[8] The DEX holds liquidity reserves that liquidity suppliers provide passively, and it defines algorithmic rules that determine execution prices and price impact. Therefore, as the block executes transactions, the price impact of each transaction is mechanical. This is in contrast to electronic markets with continuous trading, where price impact reflects the Bayesian updating of market makers who trade actively.

We define a blockchain market as the combination of a DEX and the blockchain infrastructure on which it operates. Mechanical price impact, passive liquidity provision, and paid priority are the defining features of blockchain markets that drive the results of this paper.

In this section, we introduce the features of our model for the strategic interactions of market participants in a blockchain market. A DEX supplies liquidity in a reference asset X (e.g., dollar) and in a risky asset Y. The depth of liquidity in the DEX, which we denote by L, is a summary statistic of the levels of reserves in assets X and Y, and it determines the cost of liquidity; see Appendix A.B. There are three types of agent, a representative liquidity supplier (he), informed traders (she), and noise traders with price-sensitive demand. The entry to blockchain markets, liquidity supply, and the trading process are modeled as a game that proceeds as follows.

- *Stage one*: M traders decide whether to pay a fixed information cost C to enter the blockchain and to observe private information. This choice depends on whether informed trading yields an expected future payoff higher than C.
- *Stage two*: the liquidity supplier sets the liquidity reserves in the DEX by balancing expected losses to informed traders and expected commission revenue from noise traders.

[8]Specifically, a decentralized exchange is a smart contract running on the blockchain. See Appendix A.B and Capponi and Jia (2021); Cartea, Drissi, and Monga (2024a) for more details on the mechanics of DEXs.

The level of reserves determines the cost of liquidity.

- *Stage three*: the blockchain slot starts. Informed traders observe the future liquidation value of the asset, and they value the asset at this common value plus a private adjustment. Based on valuations, traders compete for queue priority in the next block, and they determine the level of their priority fees and their trading volumes. In contrast, noise traders submit transactions with zero priority fees.[9]

After stage three, a builder constructs the block, and transactions are executed in the DEX sequentially from the highest to the lowest priority fee. Specifically, the block executes the transactions of informed traders first, followed by those from noise traders who do not pay priority fees.[10]

After the liquidity supplier sets the DEX's liquidity in stage two, we assume that he remains passive and does not compete with traders for queue priority in the subsequent block. Although the liquidity supplier does not compete directly, his strategy depends indirectly on the behavior of informed traders, because he chooses the cost of liquidity in the DEX based on rational expectations about their best response in the next stage.

In stage three, each informed trader observes the future value of the asset and a private valuation adjustment, drawn from a common and known distribution. We normalize the price of the asset at the start of the blockchain slot to zero, without loss of generality. Thus, a positive (resp. negative) future value of the asset corresponds to a positive (resp. negative) price innovation. Informed traders set priority fees and trading volumes by balancing expected trading profits against (i) the price impact generated by traders with better queue positions in the block and (ii) execution costs in the DEX.

[9]Builders are paid *gas fees* by agents to include their transactions in a block. Gas fees consist of two components, the *base fee* and the *priority fee*. The base fee is mandatory for inclusion and is paid by all traders. We assume that noise traders pay the base fee only, which we normalize to zero without loss of generality, because the base fee does not affect ordering within the block. See Appendix A.A for further institutional details on the blockchain protocol.

[10]In blockchains, informed traders do not profitably blend with noise traders to conceal their information, as they would in Kyle (1985). Blending would require paying a zero priority fee. Submitting an arbitrarily small positive priority fee is a profitable deviation, because it secures a better queue position and execution price. In equilibrium, informed traders pay non-zero priority fees which reveal valuations; see Proposition 3.

Key assumptions of the model are (i) that all informed traders know the future value of the asset, and (ii) that the sign of the price innovation determines the sign of all valuations and whether traders are buyers or sellers. We motivate these assumptions as follows. Informed traders submit their transactions near the end of the blockchain slot and rely on information observed at that time.[11] Blockchains that host decentralized markets, such as Ethereum, operate with slots of twelve seconds or longer to build blocks securely.[12] Thus, substantial price movements can occur during this period and cause the price of the asset to be far from its initial value, so most informed traders are either buyers or sellers. Our focus is on price discovery and liquidity in such settings, and the PGA is the mechanism through which information is incorporated into prices.[13]

We solve for a perfect Bayesian equilibrium of this game by backward induction. At stage three (Section II), given a known level of liquidity supply L and a known number M of competing informed traders, the priority fee bidding strategies Φ and trading volumes Q are determined. At stage two (Section III), given a known number M of competing informed traders, the liquidity supply L is determined based on the supplier's rational expectations of future trading volumes and noise trading activity. At stage one (Section IV), the number M of informed traders who pay the cost of information C is determined. The sequence of events is described in more detail in the next sections.

[11] Informed traders transact through private memory pools or wait until the end of block time in public memory pools, consistent with the empirical evidence reported in Appendix A.A.

[12] Layer-two blockchains achieve shorter block times by relying on centralized sequencers, and thus do not qualify as decentralized.

[13] In contrast, when the future value of the asset is close to its initial value, valuations can take both signs so buyers compete with sellers. In such situations, if an informed buyer and an informed seller both submit transactions within the same block, each has an incentive for the other's trade to be executed first, because this improves their own execution price. Consequently, the expected value of queue priority becomes non-positive for both traders, leading to zero equilibrium priority fees and no informational content in the auction outcome.

II. Priority fees and trading volumes

This section studies the priority fee bidding strategies and trading volumes of informed traders who compete for queue priority in stage three. Traders take as given the liquidity supply L in the DEX and the number M of competitors. Assume that the liquidation value V is positive, so informed traders wish to buy the asset. The case in which informed traders are sellers is symmetric and yields identical equilibrium priority fee bidding strategies, and trading volumes of opposite sign.

A. Assumptions

In stage three, M risk-neutral informed traders compete for queue priority in the next block. As described in Section I, transactions in a block are executed sequentially according to priority fees, and the execution costs and price impact of each transaction are determined mechanically by the DEX. In our model, we assume that the DEX offers a convex (quadratic) price schedule, and implements a linear price update rule. Specifically, let L denote the depth of liquidity in the DEX determined in stage two. A transaction with volume Q is executed with execution costs Q/L per unit of the asset, so the total cost of executing the transaction is Q^2/L. After each transaction is executed, the price updates linearly in Q, by $2\,Q/L$. This assumption is a first-order approximation of the pricing mechanism in standard DEXs; see Appendix A.B.

Traders compete to avoid potentially worse execution prices resulting from the price impact of preceding transactions in the block.[14] Thus, the supply L influences competition outcomes because it determines the magnitude of price impact. Consider two traders, 1 and 2, who sequentially buy quantities $Q_1 > 0$ and $Q_2 > 0$ of asset Y. Recall that the DEX's initial price at the start of the blockchain slot is normalized to zero. Thus, the execution price for trader 1, assuming she has queue priority, is Q_1/L, after which the DEX price

[14]When traders are risk-averse, they also compete for earlier queue positions to avoid uncertainty in execution prices.

updates to $2\,Q_1/L$. Consequently, the execution price for trader 2, who trades second in the block, is $Q_2/L + 2\,Q_1/L$.[15]

The PGA among traders is a *blind* multi-prize contest in which each competitor's payoff depends on the rank of her priority fee relative to others. The PGA is blind because informed traders do not observe each other's priority fees before submitting their own.[16] Moreover, traders' transactions are all executed and builders collect priority fees regardless of relative ranks within the block.

Before submitting their transactions, traders observe the future value V of the asset, and they hold private valuations $\{v_1, \ldots, v_M\}$. Specifically, trader i's valuation is

$$v_i = V + a_i\,, \tag{1}$$

where a_i is a private independent adjustment to the common value V which reflects trader i's hedging, inventory, risk, and other private trading motives. The adjustments $\{a_1, \ldots, a_M\}$ are independent and identically distributed (i.i.d.) across traders and centered at zero, so the future value is the average valuation, i.e., $V = \mathbb{E}\,[v_i]$.

The valuations $\{v_1, \ldots, v_M\}$ are i.i.d. with continuous cumulative distribution function F on $[0, \overline{v}]$, which admits a continuous density f.[17] Throughout, $\mathbb{E}_i$ denotes the expectation conditional on the information of trader i, which consists of the future value V and her adjustment a_i. In the perfect Bayesian equilibrium of the PGA, the M risk-neutral traders determine their priority fees $\{\Phi_1, \cdots, \Phi_M\}$ and their trading volumes $\{Q_1, \cdots, Q_M\}$.

[15] See Appendix A.B for more details on execution prices and price impact in DEXs.

[16] As discussed in Appendix A.A, the PGA is blind because traders either compete in a private memory pool or submit their transactions to the public memory pool immediately before block creation.

[17] We assume that the support of F starts at zero because the future price of the asset is large relative to (i) its initial value and (ii) private adjustments.

B. *The priority gas auction*

This section derives the expected utility of competing traders in the PGA. The contest is symmetric, and we describe it from the perspective of trader $i \in \{1, \ldots, M\}$. Trader i competes with $M-1$ other traders in the PGA. From her perspective, the $M-1$ valuations of her competitors are i.i.d. draws from the same distribution F; likewise, the $M-1$ trading volumes of her competitors are i.i.d. draws from the same distribution. We denote by $\Phi_{(j)}$, for $j \in \{1, \ldots, M-1\}$, the j-th smallest bid among the $M-1$ competitors of trader i, so that $\Phi_{(j)} < \Phi_{(j+1)}$ and $\Phi_{(M-1)}$ is the largest competing priority fee. We denote by $v_{(j)}$ and $Q_{(j)}$ the valuation and trading volume, respectively, of the competitor submitting the priority fee $\Phi_{(j)}$.

If trader i obtains priority in the block, i.e., if $\Phi_i > \Phi_{(M-1)}$, her buy transaction is executed first in the next block at the price $Q_i/L + \pi$ per unit of the asset, where π is a proportional commission paid to the DEX.[18] In this case, the trader's terminal wealth depends on the priority fee paid to the builder, the cash paid to the DEX to purchase Q_i units of the asset, the DEX commission π, the information cost C incurred in stage one, and the terminal value of her holdings:

$$W_{i,(M-1)} = \underbrace{-\Phi_i}_{\text{priority fee}} \underbrace{- Q_i \left(\frac{Q_i}{L} + \pi \right)}_{\text{cash paid to DEX}} + \underbrace{Q_i \, v_i}_{\text{terminal value of holdings}} - \underbrace{C}_{\text{information cost}} . \tag{2}$$

If trader i's priority fee Φ_i lies between the j-th and $(j+1)$-th lowest priority fees among her competitors, i.e., if

$$\Phi_{(j)} < \Phi_i < \Phi_{(j+1)} ,$$

then exactly $M-1-j$ traders, with trading volumes $\{Q_{(j+1)}, \ldots, Q_{(M-1)}\}$, are executed in the block before trader i's buy transaction. The price impact in the DEX when trading a

[18] In practice, the commission π is called a trading fee. We reserve the term fee for priority fees. We assume that $\pi < \overline{v} < \infty$ to ensure that the probability of non-zero trading volumes is positive; see Section II.D.

volume Q is $2\,Q/L$. Hence, trader i's execution price per unit of the risky asset is

$$\underbrace{\frac{Q_i}{L}}_{\text{execution cost}} + \underbrace{\frac{2}{L}\sum_{\ell=j+1}^{M-1} Q_{(\ell)}}_{\text{impact of traders with higher priority fees}} + \underbrace{\pi}_{\text{commission}} .$$

Accordingly, trader i's terminal wealth in this case is

$$W_{i,(j)} = \underbrace{-\Phi_i}_{\text{priority fee}} \underbrace{- Q_i\left(\frac{Q_i}{L} + \frac{2}{L}\Delta_{(j+1:M-1)} + \pi\right)}_{\text{cash paid to DEX}} + \underbrace{Q_i\, v_i}_{\text{terminal value of holdings}} - \underbrace{C}_{\text{information cost}} , \quad (3)$$

where, for convenience, we define

$$\Delta_{(n:m)} = \sum_{\ell=n}^{m} Q_{(\ell)}.$$

Finally, if trader i's priority fee is lower than the lowest competing fee, i.e.,

$$\Phi_i < \Phi_{(1)}\,,$$

then her terminal wealth is

$$W_{i,(0)} = -\Phi_i - Q_i\left(\frac{Q_i}{L} + \frac{2}{L}\,\Delta_{(1:M-1)} + \pi\right) + Q_i\, v_i - C\,. \quad (4)$$

Using the cases above, the expected utility of trader i can be written as

$$\begin{aligned}\mathbb{E}_i[W_i] &= \sum_{j=0}^{M-1} \mathbb{E}_i\Big[\mathbb{1}_{\Phi_{(j)}<\Phi_i<\Phi_{(j+1)}}\, W_{i,(j)}\Big] \\ &= \sum_{j=0}^{M-1} \mathbb{E}_i\left[\mathbb{1}_{\Phi_{(j)}<\Phi_i<\Phi_{(j+1)}}\left(-\Phi_i - Q_i\left(\frac{Q_i}{L} + \frac{2}{L}\,\Delta_{(j+1:M-1)} + \pi\right) + Q_i\, v_i - C\right)\right],\end{aligned}$$

with the conventions

$$\Delta_{(l:L)} = 0 \quad \text{whenever} \quad l > L, \qquad \Phi_{(M)} = \infty, \qquad \Phi_{(0)} = -\infty.$$

At stage three, trader i's valuation of the asset is v_i, and she knows the DEX supply L, the number of competing traders M, her trading volume Q_i, and her priority fee Φ_i. Trader i understands that her queue position depends on the rank of her priority fee Φ_i relative to the priority fees of her competitors. Hence, the expected wealth of trader i can be written as

$$\mathbb{E}_i\big[W_i\big] = -\Phi_i + Q_i\Big(v_i - \pi - \frac{Q_i}{L}\Big) - C - \frac{2\,Q_i}{L}\sum_{j=0}^{M-1}\mathbb{E}_i\Big[\mathbb{1}_{\{\Phi_{(j)}<\Phi_i<\Phi_{(j+1)}\}}\,\Delta_{(j+1:M-1)}\Big]. \tag{5}$$

The expected wealth (5) consists of (i) the expected profit $Q_i\,v_i$ from trading and (ii) several trading costs. These costs include the priority fee paid to builders, the commission, the information cost, the expected price impact generated by trader i's competitors, and the execution price paid to the DEX.

When submitting a transaction, trader i determines both her priority fee Φ_i and her trading volume Q_i, thus she solves the problem[19]

$$\sup_{Q_i}\sup_{\Phi_i}\mathbb{E}_i\left[W_i\right]. \tag{6}$$

Intuitively, traders buy and sell volumes proportional to their private valuations of the asset. In the optimization problem (6), the trading volume Q_i for trader i depends on v_i. Similarly, the priority fee Φ_i depends on the trading volume Q_i. Thus, in equilibrium, both the volume Q_i and the priority fee Φ_i are functions of the private valuation v_i.[20]

An equilibrium of the PGA is a mapping from valuations to a direction (buy or sell),

[19]The order of the two suprema can be interchanged.

[20]Specifically, note that $\partial_{Q_i}\mathbb{E}_i\left[W_i\right]$ is a function of v_i, i.e., $\partial_{Q_i v_i}\mathbb{E}_i\left[W_i\right] \neq 0$. Thus, the optimal trading volume $\arg\max_{Q_i}\mathbb{E}_i\left[W_i\right]$ is a function of the valuation v_i. Similarly, $\partial_{\Phi_i v_i}\mathbb{E}_i\left[W_i\right] \neq 0$, so the priority fee also depends on the private valuation.

trading volume Q_i, and priority fee Φ_i. Recall that we restricted the strategy set of traders here to only buying. In Appendix B, we motivate this restriction and argue that no robust pure-strategy equilibrium with sellers exists.

Section II.C derives the equilibrium priority fees for a fixed distribution of trading volumes, after which Section II.D derives the equilibrium trading volumes given these priority fees.

C. Priority fees

Here, we solve for the equilibrium priority fee under an arbitrary distribution of trading volumes. Let Q_j denote the random variable representing the trading volume of trader $j \neq i$ from the perspective of trader i. The volume Q_j is drawn from the interval $[0, \overline{Q}]$ according to a density function g with finite first and second moments, which may include an atom $p = G(\{0\})$ at zero volume. We denote by G the cumulative distribution function of volumes.

At zero volume, traders submit the minimum zero priority fee. Equivalently, they do not participate in the PGA, and their expected wealth equals $-C$.[21]

In the equilibrium of stage three, informed traders take the depth L as fixed and independent of their priority fees. Although the priority fee does not directly affect liquidity supply, it exerts an indirect influence. As shown in Section III, the liquidity supplier anticipates the priority fees and trading volumes of informed traders when determining the depth of the DEX's pool.

To better characterize the economic object of competition in the PGA, we obtain the next result.[22]

[21] In particular, no tie-breaking rule is required for zero volumes. Note also that a zero trading volume generates no price impact for subsequent transactions within the block.

[22] The proof is straightforward and is omitted.

LEMMA 1: *Trader i's expected terminal wealth is*

$$\mathbb{E}_i[W_i] = \mathbb{E}_i\big[W_{i,(0)}\big] + \frac{2\,Q_i}{L}\sum_{j=0}^{M-1}\mathbb{E}_i\Big[\mathbb{1}_{\{\Phi_{(j)}<\Phi_i<\Phi_{(j+1)}\}}\,\Delta_{(1:j)}\Big]\,, \tag{7}$$

where $W_{i,(0)}$ is defined in (4)*.*

The expression (7) for trader i's expected terminal wealth has the following economic interpretation. The term $\mathbb{E}_i\big[W_{i,(0)}\big]$ represents trader i's expected wealth if she is executed last in the block. By submitting a positive priority fee Φ_i, she obtains, with positive probability, the *prize*

$$\frac{2\,Q_i}{L}\sum_{j=0}^{M-1}\mathbb{E}_i\Big[\mathbb{1}_{\{\Phi_{(j)}<\Phi_i<\Phi_{(j+1)}\}}\,\Delta_{(1:j)}\Big]\,, \tag{8}$$

which represents the expected price impact that trader i avoids by improving her queue position.

The prize (8) of trader i from competing in the PGA increases with the rank of her priority fee Φ_i relative to that of her competitors. A higher rank secures an earlier queue position, so a larger volume is executed after trader i's transaction and the avoided price impact is larger. The prize (8) also increases with trading volume Q_i. The price impact of competitors worsens the execution price of every unit of the asset. Therefore the total cost due to price impact is proportional to the traded volume Q_i, and queue priority is more valuable for larger trades. Finally, the prize (8) decreases with the depth of liquidity L. A deeper DEX decreases price impact, so the loss is smaller.

Trader i expects her competitors to employ a continuous and differentiable priority fee bidding strategy $\Phi(\cdot)$ which is increasing in the trading volume Q. Therefore, the probability of submitting the $j+1$-th largest priority fee is

$$\mathbb{P}_i\left[\Phi_{(j)} < \Phi_i < \Phi_{(j+1)}\right] = \mathbb{P}_i\left[Q_{(j)} < \Phi^{-1}\left(\Phi_i\right) < Q_{(j+1)}\right]\,.$$

To determine her optimal priority fee, trader i solves the optimization problem

$$\sup_{\Phi_i} \mathbb{E}_i \left[W_i \right] , \tag{9}$$

which is equivalent to solving

$$\sup_{\Phi_i} \left\{ -\Phi_i + \frac{2\,Q_i}{L} \sum_{j=0}^{M-1} \mathbb{E}_i \left[\mathbb{1}_{\Phi_{(j)} < \Phi_i < \Phi_{(j+1)}} \Delta_{(1:j)} \right] \right\} . \tag{10}$$

That is, trader i faces a tradeoff. She chooses her priority fee to balance (i) the loss from paying the fee to the builder against (ii) the gain from the prize (8), i.e., the gain from a better execution price in the DEX.

To further analyze the optimization problem, the following lemma is useful, because it simplifies trader i's estimation of the expected prize from the PGA.

LEMMA 2: *Assume $M \geq 2$. The prize* (8) *is*

$$\frac{2\,Q_i}{L} \sum_{j=0}^{M-1} \mathbb{E}_i \left[\mathbb{1}_{\{\Phi_{(j)} < \Phi_i < \Phi_{(j+1)}\}} \Delta_{(1:j)} \right] = \frac{2\,Q_i}{L} \, (M-1) \int_0^{\Phi^{-1}(\Phi_i)} x \, dG\left(x\right) . \tag{11}$$

Lemma 2 shows that the prize is the average price impact of a competing trader when her trading volume is below trader i's inverse bid, scaled by the number $M-1$ of competitors. This result follows from (i) the independence of private adjustments, (ii) the additive and linear price impact in the DEX,[23] and (iii) the strict ordering of transactions by priority fee within the block.[24]

In a Bayesian–Nash equilibrium, trader i finds it optimal to adopt the same differentiable and increasing strategy Φ, which pins down the equilibrium strategy. The solution is characterized in the following result.

[23] In practice, nonlinearities exist in the impact of consecutive trades; see Cartea, Drissi, and Monga (2023). However, we model competition within a block whose duration is short enough for our linear approximation (A3) to be accurate.

[24] In practice, while builders have reputational incentives to maintain strict ordering, they are not required to do so by the blockchain protocol. Such strategic considerations are beyond the scope of this work.

PROPOSITION 1: *The equilibrium priority fee for a fixed distribution G of trading volumes is*

$$\Phi\left(Q_i\right) = \frac{2}{L}\left(M-1\right)\int_0^{Q_i} x^2\, dG\left(x\right). \tag{12}$$

The priority fee is increasing in the trading volume Q_i, increasing in the number of informed traders M, and decreasing in the liquidity depth L. Finally, trader i's objective (10) *is*

$$\frac{2}{L}\left(M-1\right)\int_0^{Q_i} x\,\left(Q_i - x\right)\, dG\left(x\right)\,. \tag{13}$$

To set her priority fee, trader i uses the reservation priority fee

$$\frac{2\,Q_i}{L}\left(M-1\right)\int_0^{Q_i} x\, dG(x)$$

as a baseline when computing her optimal fee. This reservation priority fee is the level that makes trader i indifferent between participating in the PGA and abstaining, that is, it sets her objective (10) equal to zero. Trader i then applies a discount to this baseline to capture a surplus. In particular, as trader i's trading volume Q_i increases, the likelihood that her reservation fee exceeds those of other traders also increases. Consequently, trader i applies a larger discount as the value of her trading volume increases.

For a fixed arbitrary distribution G of trading volumes, equilibrium priority fees inherit the properties of the PGA prize (8). Specifically, priority fees increase with trading volumes and decrease with the depth of liquidity. Moreover, Proposition 1 shows that priority fees increase with the number M of traders. From the perspective of trader i, a larger number $M-1$ of competing traders increases the expected number of traders with higher priority fees than hers. Therefore, it also increases the expected price impact of competing traders with earlier queue positions and worsens the price she obtains. This makes queue priority more valuable and increases priority fees.

D. *Trading volumes and participation cutoff*

The section above derived the equilibrium priority fees of informed traders for an arbitrary distribution G of trading volumes. This section endogenizes that distribution. Informed traders anticipate the priority fees (12) when they set their trading volumes. In particular, they anticipate that these fees increase with the volume they wish to trade (Proposition 1) and the number of competing traders.

Substituting the optimal priority fee Φ_i from (12) into the objective function (6), we express trader i's optimization problem as

$$\sup_{Q_i} \left\{ -\frac{2}{L}\,(M-1)\left(\int_0^{Q_i} x^2\,dG(x) + Q_i \int_{Q_i}^{\overline{Q}} x\,dG(x)\right) + Q_i\left(v_i - \pi - \frac{Q_i}{L}\right)\right\}. \tag{14}$$

The programme (14) is interpreted as follows. All traders employ the same priority fee bidding strategy, allowing them to estimate the expected costs of the PGA in the form of priority fees and price impact from competing traders. Additional trading costs incurred by trader i include (i) the execution cost Q_i^2/L in the DEX and (ii) the commission $\pi\,Q_i$. These costs increase with traded volume, creating an incentive for trader i to reduce her holdings Q_i. Conversely, trader i's expected trading profits are $Q_i\,v_i$. This provides an incentive for trader i to increase her holdings.

We look for equilibria in which trading volumes are a weakly monotone function of private valuations, and we write $Q_i = Q(v_i)$ for $v_i \in [0, \overline{v}]$. In particular, the minimum and maximum trading volumes satisfy

$$Q(0) = 0 \qquad \text{and} \qquad \overline{Q} = Q(\overline{v})\,.$$

Later, we verify that the monotonicity property holds in equilibrium. The distribution of trading volumes, expressed as a function of F, is given by

$$G(x) = F\big(Q^{-1}(x)\big)\,, \tag{15}$$

where $Q^{-1}(\cdot)$ denotes the generalized inverse of the function $Q(\cdot)$.

The price impact of competing traders depends on trader i's private valuation of the asset. Lower valuations correspond to lower volumes, lower volumes correspond to lower priority fees, lower priority fees correspond to worse queue positions in the block, and worse queue positions correspond to incurring higher price impact from traders with better positions. For sufficiently low valuations, the price impact associated with high-valuation traders may discourage trader i from submitting a transaction if doing so results in a negative objective (14). In what follows, we show that there indeed exists a participation cutoff $\underline{v}_M$ for private valuations below which trader i optimally chooses not to submit a transaction.

Using (15), the first-order condition (FOC) derived from the optimization problem (14) yields the following implicit integral equation for trader i's optimal trading volume

$$(v_i - \pi)\, L - 2\, Q(v_i) - 2\,(M-1) \int_{v_i}^{\overline{v}} Q(x)\, dF(x) = 0\,. \tag{16}$$

Next, Lemma 3 characterizes the participation cutoff, and Proposition 2 derives the equilibrium trading volumes and their distribution.

LEMMA 3: *Fix $M \geq 2$. There exists a unique $\underline{v}_M \in (0, \overline{v})$ which satisfies*

$$\underline{v}_M - \pi - \int_{\underline{v}_M}^{\overline{v}} \left(e^{(1-F(u))(M-1)} - 1\right) du = 0\,. \tag{17}$$

Moreover, $\underline{v}_M$ is strictly increasing in M and $\lim_{M\to\infty} \underline{v}_M = \overline{v}$.

PROPOSITION 2: *Fix $M \geq 2$. If $v_i < \underline{v}_M$, where $\underline{v}_M$ is defined in (17), then trader i's optimal trading volume is $Q(v_i) = 0$. If $v_i \geq \underline{v}_M$, then the optimal trading volume is proportional to the liquidity supply L,*

$$Q(v_i) = L\, \tilde{Q}(v_i)\,, \tag{18}$$

where

$$\tilde{Q}(v_i) = \frac{1}{2}\left((\overline{v} - \pi)\, e^{-(M-1)(1-F(v_i))} - \int_{v_i}^{\overline{v}} e^{-(M-1)(F(u)-F(v_i))}\, du \right). \tag{19}$$

Moreover,

$$Q(\underline{v}_M) = 0 \qquad \textit{and} \qquad \overline{Q} = Q(\overline{v}) = L\frac{\overline{v} - \pi}{2}.$$

The trading volume $Q(v_i)$ is increasing in the private valuation v_i, and decreasing in the number of informed traders M with limit $\lim_{M\to\infty} Q(v_i) = 0$ for every $v_i < \overline{v}$.

Proposition 2 shows that there is a participation cutoff $\underline{v}_M$ for private valuations, because informed traders with valuations below $\underline{v}_M$ find it unprofitable to trade and abstain from submitting a transaction. This result is central and drives our conclusions. It follows from the defining features of blockchain markets. Namely, liquidity is passive, price impact is mechanical rather than the result of the Bayesian updating of market makers, and paid priority determines which traders bear the price impact of others. High-valuation traders find it more profitable to bid high fees and are executed first, but the combination of mechanical price impact and the ordering of transactions in the block makes it expensive for low-valuation traders to participate.

The participation cutoff $\underline{v}_M$ strictly increases in the number of competitors M and converges to the upper bound $\overline{v}$ as $M \to \infty$. The intuition behind this result is as follows. A higher number of competing traders increases the expected volume that executes before any given trader, and therefore worsens the execution price she obtains per unit of the asset. Therefore, a higher valuation of the asset, or equivalently a higher expected profit, is required to break even.

When a trader's valuation exceeds the participation cutoff $\underline{v}_M$, Proposition 2 shows that her trading volume is proportional to the liquidity supply L. Thus, deeper liquidity reserves allow traders to extract greater profits from their information. In contrast, trading volumes decrease with the number of traders in anticipation of higher expected PGA costs.

Equilibrium trading volumes increase with the private valuation v_i. This occurs for two

reasons. One, a higher valuation means more utility from trading, and two, it increases the expected surplus from the PGA. Thus, higher private valuations correspond to larger trading volumes, and as shown in Proposition 1, they are also associated with better queue positions. We test the predictions of our model empirically in Section II.F.

Finally, the next result derives the equilibrium priority fees accounting for strategic trading volumes.

PROPOSITION 3: *The equilibrium priority fee is zero if* $v_i < \underline{v}_M$. *For* $v_i \geq \underline{v}_M$, *the priority fee, accounting for strategically adjusted trading volumes, is*

$$\Phi\left(v_i\right) = 2\,L\left(M-1\right)\int_{\underline{v}_M}^{v_i} \tilde{Q}\left(x\right)^2\,dF\left(x\right)\,. \tag{20}$$

The priority fee is increasing in the depth of liquidity L, *and, for every* $v_i < \overline{v}$, *it admits the limit* $\lim_{M\to\infty}\Phi(v_i) = 0$.

E. Competition and price formation

Proposition 2 shows that the equilibrium distribution of trading volumes is a transformation of the truncated distribution of valuations over $[\underline{v}_M, \overline{v}]$:

$$G\left(\{0\}\right) = F(\underline{v}_M) \qquad \text{and} \qquad G\big(Q(v)\big) = F(v) \quad \text{for } v \in [\underline{v}_M, \overline{v}]\,. \tag{21}$$

For a given number M of competing traders, the expected number of traders with sufficiently high valuations for the asset to submit non-zero trading volumes is $M\big(1-F(\underline{v}_M)\big)$. We refer to these traders as *active traders*.

In what follows, we assume that the probability density function of valuations f does not exceed the uniform density in a (left) neighborhood of $\overline{v}$.[25] That is, there exists $\eta > 0$ such

[25]The upper bound $\overline{v}$ on valuations can be arbitrarily large to accommodate extreme valuations and ensure our assumption holds.

that

$$f(u) \;\leq\; \frac{1}{\overline{v}-\pi} \qquad \text{for all} \quad u \in [\overline{v}-\eta, \overline{v}]. \tag{22}$$

The next result characterizes the properties of price and volume in blockchain markets as competition intensifies.

PROPOSITION 4: *The expected aggregate trading volume of active informed traders admits the closed form*

$$L\,M \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(x)\,dF(x) \;=\; \frac{L\,M\,(\underline{v}_M-\pi)}{2\,(M-1)}\,. \tag{23}$$

The aggregate trading volume admits the limit $L\,\dfrac{\overline{v}-\pi}{2} = L\,\tilde{Q}(\overline{v})$ *as competition intensifies* $(M \to \infty)$*. Moreover, the expected price at the end of the block admits the closed form*

$$\frac{M\,(\underline{v}_M-\pi)}{M-1} \tag{24}$$

and converges to $\overline{v}-\pi$*. Both limits are approached from below.*

Proposition 4 shows that, as competition intensifies, increasingly high valuations are required to participate in the PGA. For instance, when valuations are uniformly distributed on $[\pi, \overline{v}]$, the participation cutoff admits the form

$$\underline{v}_M = \overline{v} - (\overline{v}-\pi)\,\frac{\log M}{M-1}, \tag{25}$$

so the range $[\underline{v}_M, \overline{v}]$ of valuations of active traders shrinks at rate $\log M/M$. At the same time, the expected price at the end of the block in (24) increases strictly with M for $M \geq 2$. In the limit when $M \to \infty$, this price converges to $\overline{v}-\pi$, which corresponds to the upper bound of the valuation support net of the commission.[26]

The economic mechanism behind our result is endogenous selection in blockchain markets whereby only high-valuation traders remain active. Endogenous selection has direct impli-

[26]Equivalently, aggregate volume converges to the volume $Q(\overline{v})$ associated with an informed trader holding the maximal valuation $\overline{v}$.

cations for price formation. From the perspective of an uninformed observer, only non-zero priority fees and executed volumes reveal private information. As competition increases, the fraction of the valuation distribution that generates transactions shrinks. Thus, price formation is driven by an increasingly thin upper tail of the distribution, and the transaction price becomes systematically biased upward (downward if traders are selling). We test these predictions of our model in Section II.F.

As we further discuss below, an increase in block time, that is, the time required for a block to be created and markets to clear, may raise the upper bound of possible valuations $\overline{v}$ and amplify the upward price bias. These effects play a role in shaping the equilibrium supply of liquidity, which we characterize in Section III.

Finally, similar arguments to those of Sections II.B, II.C, and II.D show that our conclusions apply when traders wish to sell the asset. Specifically, the equilibrium is symmetric, i.e., a trader sells the same volume that she would buy for a valuation of equal absolute value and opposite sign.

F. Empirical evidence

To summarize, our results for the equilibrium of the PGA predict the structure of a block. Informed traders pay non-zero priority fees and noise traders pay none, so we expect the transactions of informed traders to come first in the block, followed by those of noise traders. Moreover, priority fees increase with trading volumes (Proposition 1) and trading volumes increase with valuations (Proposition 2), so we expect the highest-valuation traders to come first, followed by less aggressive traders. The informed part of the block therefore consists of transactions in the same direction with decreasing volume.

To test these predictions, Figure 1 presents an empirical analysis of trading volumes for assets whose price forms on Ethereum's Uniswap DEX, the most popular blockchain market. For such assets, the sequence of transactions within a block reflects on-chain price discovery rather than arbitrage against an external market. The figure shows that transactions near

the start of the block are predominantly in the same direction as the first transaction. Specifically, over the first 30% of the block, the probability that a transaction shares the sign of the first transaction is well above 50%. Later in the block, the expected net trading volume is close to zero, consistent with two-sided noise trading. This evidence supports the block structure predicted by our model, in which high-valuation traders obtain the earliest queue positions, followed by lower-valuation traders who trade in the same direction, and then by noise traders who trade in both directions.

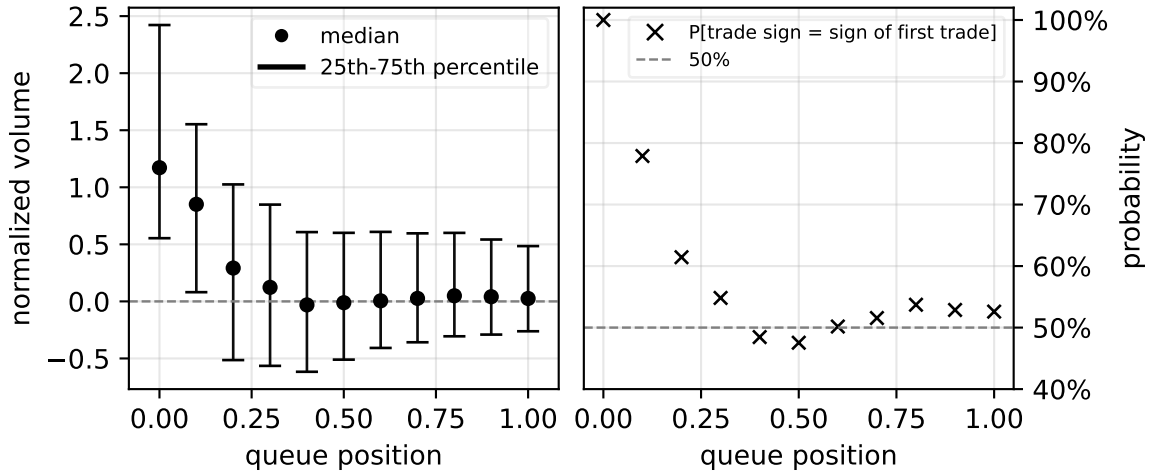


Figure 1. Left panel: median and interquartile range of (signed) trading volumes as a function of queue position within the block (0 denotes the first transaction and 1 the last). Specifically, for each asset and each block, we change the signs of observed volumes so that a positive volume denotes a transaction in the *same* direction as the first transaction of the block and a negative volume the *opposite* direction. We normalize by the block's mean absolute volume. **Right panel**: the probability that a transaction shares the sign of the first transaction of its block, as a function of queue position. The reference level of 50% corresponds to a transaction's direction being unrelated to that of the first transaction. **Data**: the data are all liquidity-taking trades between 19 May 2021 and 30 June 2026 in 75 Uniswap v3 pools on Ethereum whose risky asset is not listed on Binance, which we take to be the main venue where price formation takes place. For each asset, we keep blocks with more than five transactions and assets with more than ten such blocks. This gives 317,901 transactions in 36,692 blocks.

A natural alternative explanation for the same-direction pattern in Figure 1 is that it reflects sandwich attacks rather than informed trading. However, a sandwich attack has a different empirical signature: a front-running trade and a back-running trade in *opposite*

directions that bracket a single victim transaction, producing a round trip in volume rather than a sustained same-direction pattern.[27] The pattern in Figure 1 is inconsistent with this round-trip signature: trading volume decays monotonically in the same direction over the first 30% of the block, with no reversal. This pattern is instead consistent with the equilibrium ordering of informed traders by valuation that our model predicts.

A central prediction of our model is endogenous selection. There is a minimal valuation required for profitable informed trading, so mostly high-valuation traders submit transactions. Their trades push the price toward extreme values.

To test the prediction that paid priority biases prices toward extreme values, Table I compares the dispersion of one-block returns on the blockchain and on Binance, for a set of assets whose prices form both on-chain and off-chain (Capponi, Jia, and Yu (2026)). Specifically, we define one-block returns as price changes between two consecutive blocks that each contain at least one transaction in the asset. On the blockchain, we use the DEX price after the last transaction of each block, and on Binance, we sample prices at the timestamps of the same two blocks, so returns on the two venues are measured over identical time intervals.

The table shows that the volatility of one-block returns is higher on the blockchain than on Binance for every asset. Price moves over a block are therefore amplified on the blockchain, consistent with the endogenous selection of high-valuation traders that our model predicts.

A final consequence of endogenous selection is that, because only high-valuation traders submit transactions, the trades observed on the blockchain should be larger. This prediction is consistent with empirical evidence that per-transaction trading volumes in blockchain markets are significantly higher than those in traditional electronic limit order books; see Cartea, Drissi, and Monga (2025).

While the empirical evidence in this section establishes that the stylized facts implied by the model are present in the data, it does not, on its own, separate our mechanism from

[27]See Capponi, Jia, and Wang (2025) for an empirical characterization of sandwich attacks and other forms of maximal extractable value on Ethereum.

		Volatility (bp)	
Asset	# obs.	Blockchain	Binance
ETH/BTC	147,815	7.73	6.02
ETH	1,689,825	8.74	6.89
BTC	24,956	24.95	17.83
LINK	31,580	27.38	13.46
UNI	13,335	30.75	19.04
APE	41,730	36.41	28.67
MATIC	13,401	43.87	19.06
1INCH	3,749	84.13	34.85

Table I. Volatility of one-block returns on the blockchain and Binance. For each asset, the table reports the volatility, that is, the standard deviation of the one-block return in basis points, on the blockchain (Uniswap v3) and on Binance, between 5 May 2021 and 31 December 2023. A blockchain return is computed only between two consecutive blocks that each contain at least one transaction in the asset, and it uses the DEX price after the last transaction of each block. The corresponding Binance return uses the last Binance trade price at the timestamps of the same two blocks, so returns on the two venues are measured over the same time interval. The column "# obs." reports the number of one-block returns used for each asset. For each asset, the share of its dollar trading volume that takes place on Ethereum decentralized exchanges rather than on Binance over the same period is 29% (ETH/BTC), 40% (ETH), 4% (BTC), 11% (LINK), 19% (UNI), 14% (APE), 3% (MATIC), and 9% (1INCH). On-chain volume aggregates all Ethereum decentralized exchanges and is obtained from Dune Analytics, and the Binance figures are spot volume over the same period.

other sources of the same patterns, and we discuss these alternatives. Consider the block structure in Figure 1. First, automated liquidations by lending protocols respond to price moves and may generate multiple transactions in the same direction that are unrelated to private information. However, liquidations are concentrated in rare episodes of large price moves and their volume is small relative to DEX trading volume; see Lehar and Parlour (2022) and Chiu and Danisman (2026). Second, cross-venue arbitrage may also produce directional trading flow at the front of the block; see He et al. (2025) and Appendix A.A. However, the sample in Figure 1 consists of assets that are not listed on Binance. Moreover, cross-venue arbitrage is usually a winner-take-all competition for an observed price gap, so it predicts a single large correcting transaction followed by two-sided flow, rather than the gradual decline of volumes that we observe.

Consider next the volatility gap in Table I. First, the mechanical price impact of noise trades against thin DEX liquidity adds variance to blockchain prices even in the absence of information. However, noise flow is generally two-sided, so its expected net volume over a block with many transactions is close to zero. Second, the assets in Table I are listed on Binance, so blockchain returns may be more volatile because arbitrageurs pay priority fees to correct stale DEX prices (see Capponi and Jia (2025)). To minimize this channel, our returns are computed only between consecutive blocks that each contain at least one transaction, so corrections of prices that are stale for more than one block do not enter the sample. However, cross-venue arbitrage affects returns even between consecutive blocks. The blockchain price fluctuates around the Binance price and arbitrageurs reset these deviations from one block to the next, so part of the volatility gap in Table I may reflect channels other than selection.

III. Liquidity supply

A. Assumptions

The level of reserves deposited in the DEX determines the execution costs and the price impact of liquidity-taking trades. At stage two, a risk-neutral representative liquidity supplier sets the DEX reserves by balancing losses to informed traders with commission revenue from noise traders.[28] Liquidity supply is competitive and is determined by a zero-profit condition. Specifically, the expected commission revenue from noise traders equals expected losses to informed traders.[29]

For simplicity, we assume that noise traders transact a net volume that sums to zero

[28]Unlike traditional venues such as limit order books or dealer markets, the permissionless nature of blockchains facilitates entry to liquidity provision, particularly by less sophisticated participants. As a result, strategic liquidity suppliers cannot freely set the price of liquidity because of *noise liquidity suppliers*. This feature of blockchains may explain the losses observed in practice; see Cartea et al. (2024a).

[29]This views the representative liquidity supplier as a continuum of competitive, atomistic, price-taking liquidity suppliers operating under free entry.

in expectation, but an absolute expected volume which is price-sensitive. Specifically, the liquidity demanded by buyers is decreasing in the price $1/L$ and its expected volume is

$$\frac{N}{2} \times \frac{1}{1+\theta/L},$$

and the expected liquidity demanded by sellers is symmetric.

Demand on each side is therefore characterized by two parameters. First, $N/2$, the aggregate positive (resp. negative) liquidity needs over the blockchain slot, and second, $\theta > 0$, which governs the sensitivity of realized demand to liquidity depth L. When $L \ll \theta$, demand responds approximately linearly to liquidity, with slope proportional to $1/\theta$. As L increases, marginal gains from additional liquidity diminish. Smaller values of θ imply that most of the available demand is captured at relatively modest depth, whereas larger values imply that demand is more sensitive and requires substantial liquidity to materialize. Our specification is bounded, so realized demand does not exceed the available mass N. Liquidity demand follows, in spirit, the reduced-form models of demand in Garman (1976); Ho and Stoll (1981); Hendershott and Menkveld (2014). The liquidity supplier's expected commission revenue is therefore

$$\pi N \frac{L}{L+\theta}. \tag{26}$$

The liquidity supplier does not hold the same information about the future liquidation value V of the asset as that of informed traders. At stage two, he assumes that with probability $1/2$, the liquidation value V of the asset is positive, in which case the informed traders buy the asset and their private valuations are drawn independently from the interval $[0, \overline{v}]$ according to the density f. Similarly, with probability $1/2$, the liquidation value of the asset is negative, in which case the informed traders sell the asset and their valuations are drawn independently from the interval $[-\overline{v}, 0]$, according to the symmetric density $f(-x)$.

Recall that only traders who value the asset above the cutoff (17) submit transactions

with volumes (18). We denote the total trading volume of informed traders by

$$\Delta_M = \sum_{k=1}^{M} Q(v_k)\,. \tag{27}$$

From the perspective of the supplier, Δ_M is a random variable, whose expected value is zero, and which is assumed to be independent of the noise liquidity demand.

The liquidity supplier balances commission revenue with expected losses to informed traders. Next, we derive the expected adverse selection cost. Let X_0 and Y_0 denote the initial cash and risky asset reserves deposited by the liquidity supplier, so his initial wealth is $X_0 + Y_0\, V_0 = X_0$. Under the DEX's price schedule, the aggregate informed flow Δ_M executes at price Δ_M/L. The liquidity supplier's cash therefore changes by $\Delta_M\,(\Delta_M/L) = \Delta_M^2/L$.[30] The liquidity supplier marks his position at the end of the block, after trading unfolds, at the prevalent DEX price $2\Delta_M/L$.[31] Thus, his wealth becomes

$$X_0 \;+\; \frac{\Delta_M^2}{L} \;+\; (Y_0 - \Delta_M)\,\frac{2\Delta_M}{L},$$

and the change in wealth attributable to informed trading is

$$\frac{\Delta_M^2}{L} \;+\; (Y_0 - \Delta_M)\,\frac{2\Delta_M}{L} \;=\; -\,\frac{\Delta_M^2}{L} \;+\; \frac{2\,Y_0\,\Delta_M}{L}. \tag{28}$$

The first term on the right-hand side (RHS) of (28) is the adverse selection cost. It is non-positive because the liquidity supplier holds more reserves of the asset whose price has decreased, or less of the asset whose price has increased. The second term on the RHS of (28) is the change in value of the liquidity supplier's initial inventory. Under the prior $\mathbb{E}[\Delta_M] = 0$, the expectation of that term is zero. Hence the expected change in the wealth

[30] The change is similar whether informed traders buy or sell.

[31] Each infinitesimal liquidity supplier marks wealth at the DEX's price because the DEX is the only venue to unwind inventory. Thus, a cash-out trade executes at the marginal price.

attributable to informed trading at stage three is

$$-\frac{1}{L}\,\mathbb{E}\left[\Delta_M^2\right] = -L\,M\,S_M\,, \tag{29}$$

where we define

$$S_M = \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(u)^2\,dF(u) \;+\; \frac{(\underline{v}_M - \pi)^2}{4\,(M-1)}\,, \tag{30}$$

and the volume function $\tilde{Q}$ is defined in (18).[32]

B. Equilibrium liquidity supply

The aggregate equilibrium liquidity supply $L^\star$ satisfies a zero-profit condition, so expected commission revenue from noise traders equals expected losses to informed traders[33]

$$\pi\,N\,\frac{L^\star}{L^\star+\theta} = L^\star\,M\,S_M\,. \tag{31}$$

The next proposition derives the equilibrium DEX liquidity supply.

PROPOSITION 5: *In equilibrium, the supply of liquidity is*

$$L^\star(M) = \frac{\pi\,N}{M\,S_M} - \theta\,, \tag{32}$$

where S_M is defined in (30)*. Liquidity supply increases in the mass N of noise trading flow, and admits the limit*

$$\lim_{M\to\infty} L^\star(M) = \frac{8\,\pi\,N}{3\,(\overline{v}-\pi)^2} - \theta\,, \tag{33}$$

[32]The first term in (30) is the expected squared volume of an active trader. The second term reflects the positive correlation between any two traders' volumes. Equations (29) and (30) are derived in the proof of Proposition 5.

[33]For simplicity, the zero-profit condition omits the variance of the net noise trading flow and the commission revenue from informed traders. These two terms have opposite signs and each is economically small.

which is approached from above. The condition for markets to remain open is

$$M\, S_M \leq \frac{\pi\, N}{\theta}\,. \tag{34}$$

Trading volumes in stage three are set strategically by informed traders to be proportional to the liquidity depth L. Thus, greater liquidity provision in stage two amplifies, on average, the dollar value of informed trading profits and, consequently, the expected losses (29) of the liquidity supplier. The liquidity supplier anticipates that the dispersion S_M of aggregate trading volumes determines these losses and decreases supply accordingly. Conversely, he anticipates that noise demand is profitable and increases supply in response.

Now, we discuss how the degree of competition in the PGA affects liquidity supply in equilibrium. For simplicity, consider the case where valuations are uniform. As the number M of competing traders increases, the participation cutoff (25) strictly increases, so participation is increasingly restricted to traders with high valuations. Therefore, the aggregate trading volume (23) of informed traders and the expected price (24) at the end of the block both increase strictly with M toward their limits. As a consequence, the expected adverse selection cost per unit of supplied liquidity also increases. The liquidity supplier anticipates this and decreases DEX liquidity, which admits the form

$$L^{\star}(M) \;=\; \frac{8\,\pi\, N\,(M-1)^3}{M(\overline{v}-\pi)^2\Big[(M-1)(3\,M-5)-2(2\,M-3)\log M+2(\log M)^2\Big]} - \theta\,, \tag{35}$$

and is strictly decreasing in M. In the limit when $M \to \infty$, the price is most biased, liquidity is lowest, and only a small portion of private information is revealed.

One blockchain round corresponds to multiple short clearing rounds in a traditional electronic market with continuous trading. Therefore, our results stand in contrast to the effect of competition in traditional electronic markets. In such markets, as the number of informed traders tends to infinity, prices become fully revealing in early rounds and adverse selection vanishes in later rounds; see Holden and Subrahmanyam (1992) and Foster and

Viswanathan (1996). Market depth increases as private information is incorporated through successive rounds of trading.

This contrast arises from the defining features of blockchain markets. Trades are executed in a single clearing round, they are priced according to priority fees, and the price impact between transactions is mechanical rather than the result of the Bayesian updating of market makers. The market clears only once, so the liquidity supplier must absorb the full adverse selection risk from aggregate informed volume in a single decision, rather than adjusting liquidity across successive rounds. Price impact is mechanical and traders must pay for queue position, so the informed flow that the liquidity supplier absorbs comes mostly from high-valuation traders.

C. Market viability

Under the zero-profit condition (31), the equilibrium liquidity supply $L^\star$ is nonnegative only if the expected commission revenue from noise traders can cover expected losses to informed traders. This requires condition (34) for the viability of blockchain markets. Condition (34) is standard. It states that if the revenue from liquidity demand is less than losses to informed traders, liquidity provision is unsustainable. The degree to which this condition must hold depends on the price-sensitivity parameter θ. All else being equal, as demand becomes more sensitive to the price of liquidity, the viability of liquidity provision requires either greater liquidity demand N, a higher commission π, or reduced variance in informed trading volumes. This condition mirrors that in Glosten and Milgrom (1985) where markets shut down due to a liquidity freeze if informed traders are numerous or if the valuation of informed traders is too precise relative to the price-sensitivity of liquidity demand.

Our analysis in this section treats the number M of informed traders as exogenous. In the next section, we endogenize entry into blockchains. The equilibrium number of competing traders in the PGA is constrained by the cost of acquiring information and by the limited profitability induced by competition and defensive liquidity supply.

IV. Information acquisition

Our results in Section III show that, upon observing their private valuations, traders with low valuations submit zero trading volumes, while traders with high valuations participate in the PGA. At stage one, however, traders must decide whether to incur the information cost C to acquire private information before observing their valuations. These costs include expenditures on proprietary blockchain data, centralized exchange data, searcher contract deployment, and blockchain monitoring tools.

Assume that the total number of traders eligible to acquire information is arbitrarily large, and let M denote the subset that chooses to do so. In equilibrium, entry into informed trading is determined by the comparison between the expected profitability of informed trading and the information cost C. At stage one, the equilibrium number M of informed traders is the largest integer value such that the marginal trader's expected profit is at least the information cost. We characterize this equilibrium number in the following result.

PROPOSITION 6: *The equilibrium number of informed traders is the largest integer M such that*

$$C \;\leq\; H(M)\,, \qquad H(M) \;:=\; L^{\star}(M)\int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(v)^2\big[1-2(M-1)(1-F(v))\big]\,dF(v), \tag{36}$$

where $\underline{v}_M$ and $\tilde{Q}$ are defined in (17) *and* (19)*, respectively. The function $H(M)$ is the ex-ante trading profit per informed trader, net of execution costs and priority fees, and satisfies $\limsup_{M\to\infty} H(M) \leq 0$. If $C > 0$ and $H(2) > C$, then an equilibrium number $M^{\star} \geq 2$ of informed traders exists and is given by the largest integer M satisfying* (36)*. Moreover, $M^{\star}$ is non-decreasing in the noise mass N, and non-increasing in the information cost C.*

In the equilibrium condition (36), the integrand on the right-hand side represents the expected profit from participating in the PGA when a trader observes a valuation $v > \underline{v}_M$ and liquidity supply is set at its equilibrium level (32). These expected profits are weighted

by the likelihood $dF(v)$ of observing valuation v in stage three. Proposition 6 also shows that the mass of informed traders decreases with the cost of information and increases with the size of noise demand.

In traditional electronic markets, price efficiency is directly related to the number of traders who incur the cost of acquiring information. Each revealed valuation reduces the conditional variance of the fundamental price from the perspective of an uninformed observer. Thus, lower information costs C and greater noise trading improve price efficiency. However, under paid-priority discriminatory pricing of blockchains, price efficiency is also related to the participation cutoff (17) below which traders abstain from the PGA. An increase in entry raises the participation cutoff, so that only increasingly high valuations are revealed. Information becomes more truncated, and price efficiency deteriorates.

V. Block time

Our model endogenizes participation in the PGA, trading volumes, liquidity supply, and the amount of information in the market. The primitives are the sensitivity to prices and the total mass of noise trading flow, the distribution of private valuations, and the cost of acquiring information. An additional primitive specific to blockchains is the length of the slot, i.e., block time.

Block time is the delay traders must wait before the market clears, and it is necessary to secure the blockchain. It therefore shapes both the distribution of private valuations at clearing and the intensity of noise trading within a block. Specifically, longer blocks increase the likelihood of large innovations in prices and widen the dispersion of private valuations, and they also deteriorate execution quality for noise traders within the block.

Let T denote block time, which is known to all agents. We make the following assumptions to formalize the effect of block time on the primitives of our model.

ASSUMPTION 1: *(i) The distribution $F(\cdot\,;T)$ of private valuations has support $[0, \overline{v}(T)]$*

where $\overline{v}(T)$ *is strictly increasing in block time* T*, with* $\lim_{T\to\infty}\overline{v}(T)=\infty$*. For any* $T'>T$*,* $F(\cdot\,;T')$ *strictly first-order stochastically dominates* $F(\cdot\,;T)$ *on* $(0,\overline{v}(T)]$*, i.e.,* $F(v;T')<F(v;T)$ *for all* $v\in(0,\overline{v}(T)]$*.*

(ii) The mass of noise trading flow $N(T)$ *is decreasing in* T *and nonnegative.*

Assumption 1(i) states that longer block times widen the support of private valuations and make extreme valuations more likely. Assumption 1(ii) states that longer block times reduce the mass of noise trading flow. This reflects greater uncertainty about execution prices within longer blocks. The following corollary presents the comparative statics implied by these assumptions and the results derived in the previous sections.

COROLLARY 1: *Fix* $M\geq 2$ *and let Assumption 1 hold. The participation cutoff* $\underline{v}_M(T)$ *in* (17) *is increasing in* T*. With many informed traders (*$M\to\infty$*), the DEX price* $\overline{v}(T)-\pi$ *at the end of the block is increasing in* T*, equilibrium liquidity is decreasing in* T*, and markets shut down for sufficiently large* T*.*

Corollary 1 shows that longer block times discourage participation in informed trading and push prices to more extreme levels, which impairs price discovery. The result also shows that, when many traders compete, longer blocks reduce liquidity, and beyond a critical block time, the market collapses.

Longer block times reduce noise demand, by Assumption 1(ii). Holding the valuation distribution fixed, this channel lowers DEX liquidity for any fixed number of informed traders, tightens the market viability condition (34), and reduces the number of informed traders $M^\star$. Our results show that although longer block times enhance blockchain security, they decrease liquidity, deter entry, and impair price efficiency.

VI. Conclusions

The stated ambition of blockchains is to provide a decentralized alternative to intermediaries and to organize economic activity at scale. Financial markets are a central component

of that ambition. We proposed a theoretical model of price formation and liquidity in a blockchain market based on paid-priority queues and discrete clearing.

Our analysis showed that the very features that guarantee decentralization, namely block time to conduct consensus and priority fees to compensate block builders, may undermine market viability. If blockchains are to become a credible infrastructure for price discovery, protocol design must carefully address how transactions are organized and prioritized within each block.

Appendix A. Institutional details

The microstructure of DEXs differs fundamentally from that of traditional financial markets due to the unique infrastructure on which they operate. In this section, we outline some institutional features of blockchains that underpin the framework of our model.

A. Ethereum, memory pools, and priority gas auctions

The Ethereum blockchain is a distributed and public digital ledger stored by *nodes*. Blocks of transactions are sequentially added to the ledger at regular intervals called block time. Block time must be long enough to allow for secure confirmation of blocks because this process requires aggregating a large number of cryptographic signatures in multiple rounds of network communications. Shortening block time excludes slower participants and concentrates rewards among those with superior connectivity, which would undermine decentralization. For example, block time on Ethereum is deterministic and fixed at twelve seconds.

When an agent submits a transaction, it is not immediately included in a block. Instead, the pending transaction is stored in a memory pool until the end of block time. For a pending transaction to be executed, it must be included in a block created by a builder. Traders attach gas fees to transactions. Gas fees consist of a mandatory base fee and an optional priority fee. The base fee is burned, i.e., it is removed from circulation, and builders, among other revenue sources (Drissi, Feinstein, and Williams (2026)), are paid the priority fees by traders to secure the blockchain. The priority fee incentivizes quick inclusion and priority in the block. Thus, agents compete in PGAs for better execution prices or to exploit arbitrage opportunities.

Accordingly, there are three stages in the lifecycle of a transaction: (i) *submission*, where an agent specifies the transaction details and sets a gas fee; (ii) *storage in the memory pool*, where the pending transaction awaits selection by a builder; and (iii) *inclusion in a block*,

which corresponds to the execution of the transaction. Transaction submission may occur at any time within the blockchain slot, and it is executed provided the transaction reaches the builder's memory pool before block creation.[34]

Memory pools and PGAs. There are two types of memory pools, public and private. Pending transactions in the public memory pool are visible prior to execution, so agents operating on the Ethereum blockchain are exposed to arbitrage and predatory attacks. Private memory pools have recently emerged in the blockchain ecosystem to protect users; see Capponi et al. (2025); Wang, Huang, Zhang, Huang, Wang, and Tang (2025).[35] Agents in these private channels send their transactions directly to builders to conceal transaction details and avoid attacks. At present, the vast majority of trading flow in Ethereum DEXs passes through private memory pools.[36] For additional details on blockchain protocols, see John, Monnot, Mueller, Saleh, and Schwarz-Schilling (2025).

Next, we argue that our model is suitable for PGAs among informed traders in both types of memory pools when the blockchain is the sole venue for price formation.

PGAs differ across public and private memory pools. In private memory pools, traders do not observe each other's bids before block creation, so the PGA is sealed-bid as described in our model above. In contrast, in public memory pools, bids are visible prior to execution, and traders compete in an online auction throughout the blockchain slot.[37]

In public memory pools, informed traders benefit from waiting until the end of block time to submit their transactions for three reasons. First, early bids reveal information to competitors, second, early bids set a minimum fee due to the *account nonce*;[38] and third, late

[34] In practice, if there are too many transactions in the memory pool, only a fraction of all transactions is executed.

[35] www.flashbots.net

[36] See www.dune.com/queries/5184076/8532375.

[37] Another difference between the two types of memory pools is that in public memory pools, fees for failed transactions are paid regardless of execution, whereas in private memory pools, transactions that are not executed do not pay priority fees.

[38] A *nonce* tracks the number of transactions and it is assigned to each user to prevent replay and double-spending attacks; see Vujičić, Jagodić, and Ranđić (2018). Users can only modify pending transactions by submitting a new one with the same nonce and a higher gas fee.

bidding allows users to incorporate the most up-to-date information. As a result, late bidding has emerged as the dominant strategy in public memory pools. We study the distribution of transaction submission time in the public memory pool of Ethereum in Figure 2 to support this claim empirically. Therefore, our model is suitable to model PGAs in public memory pools.

At present, PGAs are mainly between specialized users (searchers or bots) who compete

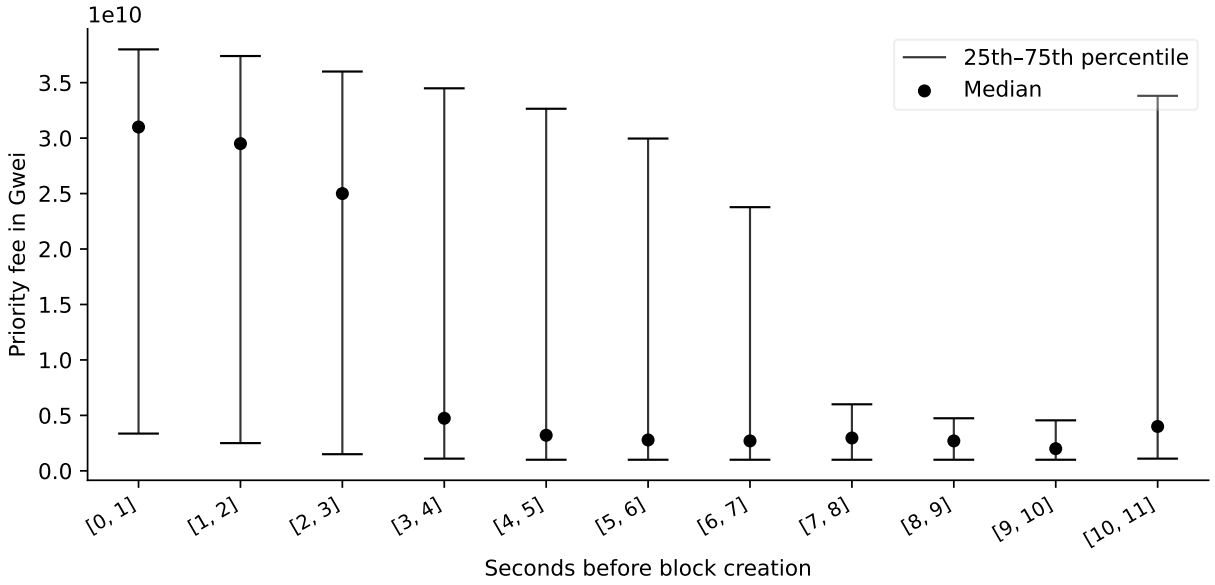


Figure 2. Distribution of priority fees as a function of timing within the blockchain slot. Informed traders are typically associated with higher priority fees, as shown empirically in Capponi et al. (2026) and theoretically in our model. The figure shows that they tend to submit their transactions very close to the end of the blockchain slot. The data are from *EthPandaOps* and it includes 10^7 transactions observed in the public memory pool of Ethereum between 20 March 2024, 00:00, and 21 March 2024, 17:08. This data corresponds to a period when the public memory pool accounted for approximately 60% of DEX trading volumes. At present (November 2025), this share is about 16%. The SQL data tables and other details are here. See Liu et al. (2022); Mizrach and Yoshida (2025) for a more detailed empirical analysis.

to exploit the discrete-time structure of blockchains through two types of strategies. First, attacks on transactions in public memory pools,[39] and second, arbitrage opportunities across DEXs or between a DEX and a competing centralized venue such as Binance; see Heimbach,

[39]These include frontrunning, backrunning, and sandwich attacks.

Pahari, and Schertenleib (2024). Effectively, this type of PGA is a first-price sealed-bid auction in which the value of the item is public and known to all competitors. As predicted by standard theory and as observed empirically, the majority of the revenue from such PGAs is ultimately paid back to builders in priority fees. The objective of this paper is not to study PGAs for arbitrage. Instead, we focus on market outcomes when PGAs are the means for price formation and private information is incomplete.

Sequential market clearing. In our model, we make the simplifying assumption that builders strictly sort transactions in the block only by priority fee. In practice, they are not required by the blockchain protocol to do so. However, failing to do so harms reputation and weakens agents' incentive to pay priority fees. Moreover, the Ethereum Foundation is actively improving the blockchain protocol to impose rules for the behavior of builders to reduce censorship and rent extraction; see Thiery, D'Amato, Ma, Monnot, Tsao, Kaufmann, and Song (2024); Wadhwa, Ma, Thiery, Monnot, Zanolini, Zhang, and Nayak (2025).

B. Automated market makers

In this section, we show how the market frictions assumed in our model above apply to automated market makers (AMMs). In AMMs, algorithmic rules specify how liquidity supply, in the form of aggregated reserves in a liquidity pool, determines execution prices and price impact; see Lehar and Parlour (2025), Capponi and Jia (2021), and Cartea et al. (2025).

Assume an AMM that supplies liquidity x in the reference asset X and y in the risky asset Y. A convex trading function Γ defines the combinations of reserves $\{x = \Gamma(y), y\}$ that make liquidity suppliers indifferent, i.e., an iso-liquidity curve. Let $Q > 0$ denote a quantity of Y that a trader wishes to buy (resp. sell). The trader pays (resp. receives) $\Gamma(y-Q)-\Gamma(y)$

(resp. $\Gamma(y+Q)-\Gamma(y)$) in asset X. Thus, execution prices per unit of asset Y are

$$\text{price to sell } Q: \quad \frac{\Gamma(y)-\Gamma(y+Q)}{Q}\,, \qquad \text{price to buy } Q: \quad \frac{\Gamma(y-Q)-\Gamma(y)}{Q}\,. \tag{A1}$$

Moreover, execution prices for an infinitesimal volume $Q \to 0$ converge to $V = -\Gamma'(y)$, which is referred to as the *marginal price*. The difference between execution and marginal prices is the execution costs, and they are determined by the reserves y. The reserves also determine the impact $-\Gamma'(y \pm Q) - V$ of a trade on the marginal price.

Our model assumes a convex (quadratic) price schedule and a linear price update rule in the DEX. These are nonlinear in an AMM. Consider the following approximations of execution prices:

$$\frac{\Gamma(y)-\Gamma(y+Q)}{Q} \approx V - \frac{Q\,\Gamma''(y)}{2}\,, \qquad \frac{\Gamma(y-Q)-\Gamma(y)}{Q} \approx V + \frac{Q\,\Gamma''(y)}{2}\,, \tag{A2}$$

and price impact:

$$-\Gamma'(y+Q) - V \approx -Q\,\Gamma''(y)\,, \qquad -\Gamma'(y-Q) - V \approx Q\,\Gamma''(y)\,. \tag{A3}$$

The depth L of the DEX in our model depends on the convexity $\Gamma''(y)$ of the AMM's trading function in (A2) and (A3):[40]

$$L = 2/\Gamma''(y). \tag{A4}$$

The convexity of trading functions is a key determinant of liquidity; see Milionis et al. (2022); Cartea et al. (2023). Akin to traditional electronic markets, market frictions in AMMs increase with the traded volume Q and decrease with the depth L. Figure 3 uses DEX data from the Ethereum blockchain to show that approximating the AMM's pricing

[40] For example, the trading function for constant product markets like Uniswap, which are the most popular DEXs, is $\Gamma(y) = L^2/y$, where $L = \sqrt{x\,y}$ is the pool's liquidity parameter, so $\Gamma''(y) = 2\,V^{3/2}/L$, and by (A4) the model's depth is $L = 2/\Gamma''(y) = L/V^{3/2} = y^2/x$.

rule with a linear schedule is accurate in practice because trading volumes represent a small fraction of total liquidity supply.

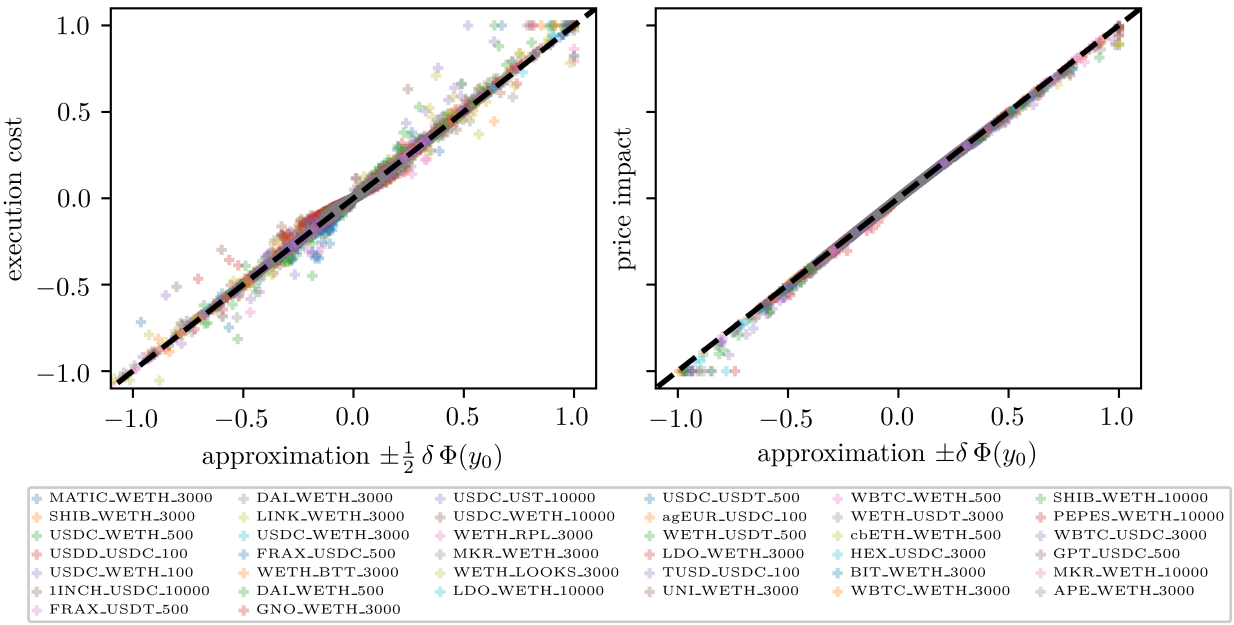


Figure 3. Scatter plots of the execution costs and the price impact of 2.622 million liquidity-taking transactions against the approximations (A2) and (A3). The transactions are between 1 January 2023 and 31 December 2023 in 38 different Uniswap v3 pools. For each pool, the execution costs and price impacts are scaled between $[0, 1]$ for buy transactions and $[-1, 0]$ for sell transactions.

Appendix B. Two-sided trading

Assume, as in the main text, that the price innovation is positive; the negative case is symmetric. In the model of Sections I and II, we restrict the strategy set of informed traders to buying, so all traders take the same side.

A trader with a low valuation might prefer to sell. Traders with high valuations trade large volumes and pay high priority fees, so they execute at the top of the block and move the price substantially. A trader with a low valuation, anticipating this, may earn more by selling the asset at the resulting high price than by buying it. This appendix motivates

the restriction of the strategy set to only buying. We argue that no robust pure-strategy equilibrium with sellers exists.

Consider the unrestricted PGA in which each informed trader chooses a direction (buy or sell), a trading volume, and a priority fee, so a pure strategy maps each trader's valuation to such a triple.[41] Noise traders behave as in Section III, and their expected net volume is zero. A candidate mapping is not an equilibrium if a trader with some valuation can change her triple and strictly increase her expected payoff.

We first show that no seller pays a positive priority fee in equilibrium. Consider a candidate equilibrium in which some traders sell and pay positive priority fees, and let $\varphi > 0$ denote the lowest priority fee paid by a seller. Suppose that no other trader pays the fee φ. Every trader with a priority fee in $(0, \varphi)$ is therefore a buyer. If the seller with fee φ deviates to a smaller fee $\epsilon \in (0, \varphi)$, she saves the fee difference $\varphi - \epsilon$, and her execution price can only improve. The deviation is strictly profitable, so the candidate is not an equilibrium.

Another candidate equilibrium in which sellers pay positive priority fees is one in which several traders, including sellers, pay the same fee φ and are ordered randomly. If the expected net volume of such a group is negative (resp. positive),[42] then a seller (resp. buyer) in the group who increases her fee infinitesimally executes before the group and improves her expected execution price in proportion to the group's expected net volume. If the expected net volume of the group is zero, a seller in the group gains by deviating to a smaller fee $\epsilon \in (0, \varphi)$ without worsening her expected execution price. Therefore, in any equilibrium, sellers pay no priority fee.

By the same arguments, there is no equilibrium in which a group of traders paying a zero priority fee has a nonzero expected net volume.

The arguments above rule out all but one class of candidate equilibria with sellers, in

[41] Consistent with the main text, we restrict attention to symmetric profiles, in which all informed traders use the same mapping from valuations to actions.

[42] The expected net volume of a group computed from the perspective of a member excludes that member's own volume. Under symmetry, each informed trader in the group has the same expected volume, so the expected net volume of the group computed from the perspective of any member has the same sign as that of the full group.

which a group of traders with zero expected net volume, including sellers, buyers, and noise traders, pays zero priority fees and executes in random order. The deviations above are no longer profitable.[43]

This candidate equilibrium is a knife-edge construction. It requires a group of buyers, sellers, and noise traders to coordinate on a zero priority fee, and it requires their expected net volume to be exactly zero. Moreover, the execution order within the group is random, and, in practice, the group would consist of hundreds of transactions. We do not view the constructed candidate equilibrium as economically robust, because it fails under arbitrarily small perturbations of the baseline environment. One such perturbation is a small aversion to uncertainty in execution prices. As noted in Section II, traders compete for queue position not only to avoid price impact but also to avoid this uncertainty. A trader in the group who pays an arbitrarily small fee reduces the uncertainty of her execution price without affecting her expected execution price. This is a strictly profitable deviation for a trader with any aversion, however small, to that uncertainty. The preference for less uncertain execution prices does not follow from the risk-neutral payoffs of our baseline model. We use this argument as a selection criterion to set aside a candidate equilibrium that is fragile to perturbations of the environment.

We therefore restrict attention to the one-sided, buy-only equilibrium characterized in the main text. This is also the equilibrium we observe in the data. As described in Section II.F, trading volume within blocks is concentrated at the front, is in the direction of the block's first trade, and decays to zero toward the back with no reversal (Figure 1). Other equilibria, including in mixed strategies, may exist, but we do not pursue them, both because the knife-edge candidate above is not robust and because it is not the equilibrium supported by the data.

[43] Under risk neutrality, this candidate may be an equilibrium. We do not prove that it is.

Appendix C. Proofs

A. *Proof of Lemma 2*

Fix trader $i \in \{1, \ldots, M\}$. G is continuous on $(0, \overline{Q}]$, so the events

$$\{Q_{(j)} < \Phi^{-1}(\Phi_i) < Q_{(j+1)}\}, \qquad j \in \{0, \ldots, M-1\}.$$

form an almost-sure partition of the sample space. The boundary cases are $\{\Phi^{-1}(\Phi_i) < Q_{(1)}\}$ for $j = 0$ and $\{Q_{(M-1)} < \Phi^{-1}(\Phi_i)\}$ for $j = M-1$.

Whenever $Q_{(j)} < \Phi^{-1}(\Phi_i) < Q_{(j+1)}$, exactly j of the $M-1$ competitor volumes fall below the inverse bid $\Phi^{-1}(\Phi_i)$, and these are precisely the volumes entering the partial sum $\Delta_{(1:j)}$. Therefore, on this event, we write

$$\Delta_{(1:j)} = \sum_{k=1}^{M-1} Q_k \, \mathbb{1}_{\{Q_k < \Phi^{-1}(\Phi_i)\}},$$

where the right-hand side does not depend on j. Multiplying by the corresponding indicator and summing over j, the partition property gives

$$\sum_{j=0}^{M-1} \mathbb{1}_{\{Q_{(j)} < \Phi^{-1}(\Phi_i) < Q_{(j+1)}\}} \Delta_{(1:j)} = \sum_{k=1}^{M-1} Q_k \, \mathbb{1}_{\{Q_k < \Phi^{-1}(\Phi_i)\}}.$$

Taking expectations and using that $\{Q_1, \ldots, Q_{M-1}\}$ are i.i.d. with distribution G, we obtain

$$\begin{aligned}\sum_{j=0}^{M-1} \mathbb{E}_i \left[\mathbb{1}_{\Phi_{(j)} < \Phi_i < \Phi_{(j+1)}} \Delta_{(1:j)} \right] &= (M-1)\, \mathbb{E}\left[Q \, \mathbb{1}_{\{Q < \Phi^{-1}(\Phi_i)\}} \right] \\ &= (M-1) \int_0^{\Phi^{-1}(\Phi_i)} x \, dG(x).\end{aligned}$$

□

B. *Proof of Proposition 1*

First, consider the function

$$\mathbb{E}\left[Q \mid Q > x\right] = \tau\left(x\right) ,$$

for the random variable Q with continuous probability distribution function g. Write

$$\tau\left(x\right) = \frac{\int_x^{\overline{Q}} u\, g\left(u\right)\, du}{1 - G\left(x\right)} ,$$

to obtain

$$\begin{aligned}
\tau'\left(x\right) =& \frac{-x\, g\left(x\right)}{1 - G\left(x\right)} + g\left(x\right) \frac{\int_x^{\overline{Q}} u\, g\left(u\right)\, du}{\left(1 - G\left(x\right)\right)^2} \\
=& \frac{g\left(x\right)}{1 - G\left(x\right)} \left(\tau\left(x\right) - x\right) .
\end{aligned}$$

Thus, the FOC derived from the optimization problem in (10) is

$$\begin{aligned}
\Phi'\left(\Phi^{-1}\left(\Phi_i\right)\right) =& \left(2\, Q_i \Big/ L\right) \left(M - 1\right) g\left(\Phi^{-1}\left(\Phi_i\right)\right) \tau\left(\Phi^{-1}\left(\Phi_i\right)\right) \\
& - \left(2\, Q_i \Big/ L\right) \left(M - 1\right) \left(1 - G\left(\Phi^{-1}\left(\Phi_i\right)\right)\right) \tau'\left(\Phi^{-1}\left(\Phi_i\right)\right) \\
=& \frac{2\, Q_i}{L} \left(M - 1\right) g\left(\Phi^{-1}\left(\Phi_i\right)\right) \mathbb{E}\left[Q \mid Q > \Phi^{-1}\left(\Phi_i\right)\right] \\
& - \frac{2\, Q_i}{L} \left(M - 1\right) g\left(\Phi^{-1}\left(\Phi_i\right)\right) \left(\mathbb{E}\left[Q \mid Q > \Phi^{-1}\left(\Phi_i\right)\right] - \Phi^{-1}\left(\Phi_i\right)\right) \\
=& \frac{2\, Q_i}{L} \left(M - 1\right) \Phi^{-1}\left(\Phi_i\right) g\left(\Phi^{-1}\left(\Phi_i\right)\right) .
\end{aligned}$$

In equilibrium, trader i finds it optimal to adopt the same strategy Φ, so $\Phi^{-1}\left(\Phi_i\right) = Q_i$, and the FOC becomes

$$\Phi'\left(Q_i\right) = \frac{2}{L} \left(M - 1\right) Q_i^2\, g\left(Q_i\right) ,$$

with boundary condition $\Phi(0) = 0$, which establishes (12). Substituting $\Phi_i = \Phi(Q_i)$ into (10)

and applying Lemma 2 with $\Phi^{-1}(\Phi_i) = Q_i$ gives

$$-\Phi(Q_i) + \frac{2\,Q_i}{L}(M-1)\int_0^{Q_i} x\,dG(x) = \frac{2}{L}(M-1)\int_0^{Q_i} x(Q_i - x)\,dG(x),$$

which establishes (13).

Next, we verify global optimality. Reparametrize the objective (10) in the bid $x^\star = \Phi^{-1}(\Phi_i)$ to obtain

$$U(x^\star) = -\Phi(x^\star) + \frac{2Q_i}{L}(M-1)\int_0^{x^\star} x\,dG(x).$$

Use the closed form

$$\Phi'(y) = (2/L)(M-1)\,y^2\,g(y)$$

derived above, to obtain

$$U'(x^\star) \;=\; -\Phi'(x^\star) + \frac{2Q_i}{L}(M-1)\,x^\star g(x^\star) \;=\; \frac{2(M-1)}{L}\,x^\star\,g(x^\star)\,(Q_i - x^\star)\,,$$

which is strictly positive on $(0, Q_i)$ and strictly negative on $(Q_i, \overline{Q}]$ whenever $g > 0$. Bidding $\Phi_i > \Phi(\overline{Q})$ wins against all competitors with probability one and gives the payoff

$$-\Phi_i + \frac{2Q_i}{L}(M-1)\int_0^{\overline{Q}} x\,dG(x).$$

The payoff is strictly decreasing in Φ_i, so it is dominated by $\Phi(\overline{Q})$. Therefore, $x^\star = Q_i$ is the unique global maximizer and the FOC's solution characterizes the global optimum. □

C. Proof of Lemma 3

For $M \geq 2$ and $v \in [0, \overline{v}]$, define the function

$$h_M(v) = v - \pi - \int_v^{\overline{v}} \left(e^{(1-F(u))(M-1)} - 1\right) du, \qquad v \in [0, \overline{v}]. \tag{C1}$$

First, we show that the function h_M is increasing in v. Note that the integrand

$$e^{(1-F(u))(M-1)} - 1$$

is continuous and nonnegative. Thus, h_M is continuous and differentiable on $(0, \overline{v})$ with derivative

$$h'_M(v) = e^{(1-F(v))(M-1)} > 1.$$

Thus h_M is strictly increasing.

Next we show the existence and uniqueness of the root. h_M is strictly increasing, so it can have at most one root. Use the endpoint conditions

$$h_M(0) < 0 \qquad \text{and} \qquad h_M(\overline{v}) = \overline{v} - \pi > 0\,.$$

By continuity of h_M, we obtain existence of a unique $\underline{v}_M \in (0, \overline{v})$ satisfying $h_M(\underline{v}_M) = 0$. Since $h_M(\overline{v}) > 0$, the root cannot equal $\overline{v}$ and thus $\underline{v}_M < \overline{v}$.

Next, we show that the root, i.e., the valuation cutoff, is increasing in the number of informed traders M that paid the cost of acquiring information in stage one. For fixed v, differentiate $h_M(v)$ with respect to M to obtain:

$$\partial_M h_M(v) = -\int_v^{\overline{v}} (1 - F(u))\, e^{(1-F(u))(M-1)}\, du < 0.$$

Thus for every v, the function $M \mapsto h_M(v)$ is strictly decreasing. Use the identity $h_M(\underline{v}_M) = 0$ and the implicit function theorem[44] to write

$$\partial_M \underline{v}_M = -\frac{\partial_M h_M(\underline{v}_M)}{\partial_v h_M(\underline{v}_M)} > 0,$$

[44]Although M takes integer values, $h_M(v)$ is well-defined for any real $M \geq 2$, so the implicit function theorem applies. Equivalently, for integer M, $h_{M+1}(\underline{v}_M) < h_M(\underline{v}_M) = 0$ because the integrand in (C1) is strictly increasing in M on the set where $F < 1$. Combined with the fact that h_{M+1} is strictly increasing in v, this gives $\underline{v}_{M+1} > \underline{v}_M$.

because $\partial_M h_M(\underline{v}_M) < 0$ and $h'_M(\underline{v}_M) > 0$. Therefore, $\underline{v}_M$ is strictly increasing in M.

Finally, we show that $\lim_{M\to\infty} \underline{v}_M = \overline{v}$. First, we use (C1) to obtain that for all $v < \overline{v}$,[45]

$$\lim_{M\to\infty} h_M(v) = -\infty.$$

Thus, for each $\epsilon > 0$, there exists a large enough $M(\epsilon)$ such that $h_M(\overline{v} - \epsilon) < 0$ for all $M > M(\epsilon)$. Use that (i) $h_M(\cdot)$ is increasing, (ii) $h_M(\overline{v} - \epsilon) < 0$, and (iii) the definition of the cutoff $h_M(\underline{v}_M) = 0$, to obtain that

$$\overline{v} - \epsilon \leq \underline{v}_M \qquad \text{for all} \quad \epsilon > 0 \ \text{ and } \ M > M(\epsilon).$$

We showed earlier that for all M, $\underline{v}_M \leq \overline{v}$. Thus, for all $M > M(\epsilon)$,

$$\overline{v} - \epsilon \leq \underline{v}_M \leq \overline{v},$$

and we conclude that $\lim_{M\to\infty} \underline{v}_M = \overline{v}$. □

D. Proof of Proposition 2

Substitute the optimal priority fee (12) in the objective (14) to write the expected wealth of trader i as

$$\mathbb{E}_i[W_i] = -\frac{2}{L}(M-1)\left(\int_0^{Q_i} x^2\, dG(x) + Q_i \int_{Q_i}^{\overline{Q}} x\, dG(x)\right) + Q_i\left(v_i - \pi - \frac{Q_i}{L}\right) - C\,.$$

The optimal trading volume is the solution of $\partial_{Q_i}\mathbb{E}[W_i] = 0$ or

$$0 = (v_i - \pi)L - 2\,Q_i - 2\,(M-1)\int_{Q_i}^{\overline{Q}} x\, dG(x)\,. \tag{C2}$$

[45] For $u \in [v, \overline{v})$, $1 - F(u) > 0$, so $e^{(1-F(u))(M-1)} \to +\infty$ pointwise and the integrand in (C1) diverges to $+\infty$ on a set of positive Lebesgue measure. By monotone convergence, $\int_v^{\overline{v}} \left(e^{(1-F(u))(M-1)} - 1\right) du \to +\infty$ as $M \to \infty$, hence $h_M(v) \to -\infty$.

Differentiate (C2) in Q_i to obtain the second-order condition

$$\partial^2_{Q_i}\mathbb{E}_i\left[W_i\right] \;=\; \frac{2}{L}\big[(M-1)\,Q_i\,g(Q_i)-1\big]\,.$$

Under the equilibrium relation $G(Q(v)) = F(v)$, the strategy $Q : [\underline{v}_M, \overline{v}] \to [0, \overline{Q}]$ is a strict bijection (established below), so every $Q_i \in (0, \overline{Q}]$ admits a unique inverse $v = Q^{-1}(Q_i)$. Differentiate to obtain

$$g(Q_i) = f(v)/Q'(v),$$

and the ODE (C3) established below implies

$$(M-1)\,Q_i\,g(Q_i) \;=\; \frac{(M-1)\,Q(v)\,f(v)}{L/2+(M-1)\,Q(v)\,f(v)} \;<\; 1\,,$$

so $\partial^2_{Q_i}\mathbb{E}_i\left[W_i\right] < 0$ for every $Q_i \in (0, \overline{Q}]$, and also at $Q_i = 0$, because $Q_i g(Q_i) = 0$ for $Q_i = 0$. Therefore, the objective is strictly concave on $[0, \overline{Q}]$.

For $Q_i > \overline{Q}$, we have

$$g(Q_i) = 0 \qquad \text{and} \qquad \partial_{Q_i}\mathbb{E}_i\left[W_i\right] = v_i - \pi - 2Q_i/L \le v_i - \overline{v} \le 0,$$

so the objective is strictly decreasing on $[\overline{Q}, \infty)$. Therefore, the solution to the FOC (C2) is the unique global maximizer.

Recall that, in equilibrium, G is the distribution of trading volumes pinned down by the increasing trading volume function Q, so

$$G\left(Q_i\right) = F\left(v_i\right)\,.$$

Rearrange (C2) to obtain the boundary condition

$$\overline{Q} = L\frac{\overline{v}-\pi}{2},$$

and differentiate to find that the trading volumes solve the ODE

$$Q'(v_i) = \frac{L}{2} + (M-1)\, Q(v_i)\; f(v_i)\,, \tag{C3}$$

whose solution is in the statement of the result. The solution is increasing in v, and trader i's optimization problem is over nonnegative volumes. Thus the boundary condition is at the lowest valuation $\underline{v}_M$ for which the trader optimally chooses a nonnegative volume. This cutoff is characterized in Lemma 3 and corresponds to when the FOC (C2) is satisfied at $\underline{v}_M$ with $Q(\underline{v}_M) = 0$. For $v_i < \underline{v}_M$, evaluate the partial derivative of the objective at $Q_i = 0$ to obtain

$$\partial_{Q_i} \mathbb{E}_i [W_i] \Big|_{Q_i=0} = (v_i - \pi) L - 2(M-1) \int_0^{\overline{Q}} x\, dG(x) = (v_i - \underline{v}_M) L < 0\,,$$

where the second equality uses (17) when $Q_i = 0$ and $v_i = \underline{v}_M$. By strict concavity of the objective on $[0, \overline{Q}]$, the constrained maximum on $Q_i \geq 0$ is at $Q_i = 0$, hence $Q(v_i) = 0$. To obtain the form in Lemma 3, set $\tilde{Q}(\underline{v}_M) = 0$ in (19), which gives

$$(\overline{v} - \pi)\, e^{-(M-1)(1-F(\underline{v}_M))} \;=\; \int_{\underline{v}_M}^{\overline{v}} e^{-(M-1)(F(u)-F(\underline{v}_M))}\, du\,.$$

Multiply both sides by $e^{(M-1)(1-F(\underline{v}_M))}$ to obtain

$$\overline{v} - \pi \;=\; \int_{\underline{v}_M}^{\overline{v}} e^{(M-1)(1-F(u))}\, du\,,$$

which, after adding and subtracting $\overline{v} - \underline{v}_M$, is equivalent to (17).

It remains to establish the monotonicity and limit properties in the statement. From the ODE (C3), we have

$$Q'(v_i) = L/2 + (M-1) Q(v_i)\, f(v_i) \geq L/2 > 0$$

on $[\underline{v}_M, \overline{v}]$, so Q is strictly increasing in v_i. Next, (19) rewrites as

$$\tilde{Q}(v_i) = \frac{e^{-(M-1)(1-F(v_i))}}{2}\left[(\overline{v}-\pi) - \int_{v_i}^{\overline{v}} e^{(M-1)(1-F(u))}\,du\right], \tag{C4}$$

where the coefficient $\frac{e^{-(M-1)(1-F(v_i))}}{2}$ in (C4) is positive and decreasing in M because $1 - F(v_i) \geq 0$. The term in brackets in (C4) is decreasing in M because the integrand $e^{(M-1)(1-F(u))}$ increases in M. The term in brackets in (C4) is also nonnegative on $[\underline{v}_M, \overline{v}]$ because the term is zero at $v_i = \underline{v}_M$ by (17), and it is increasing in v_i (the integrand is positive). Hence $Q(v_i) = L\,\tilde{Q}(v_i)$ is decreasing in M. Finally, bound the bracket above by $\overline{v} - \pi$ to obtain

$$\tilde{Q}(v_i) \;\leq\; \frac{\overline{v}-\pi}{2}\,e^{-(M-1)(1-F(v_i))},$$

which converges to zero as $M \to \infty$ for every $v_i < \overline{v}$. □

E. Proof of Proposition 3

Trader i uses the priority fee (12) for an arbitrary distribution of volumes with cumulative distribution function G:

$$\Phi\left(Q_i\right) = \frac{2}{L}\left(M-1\right)\int_0^{Q_i} u^2\,dG\left(u\right). \tag{C5}$$

The optimal volumes of PGA competitors are distributed according to the distribution G described in (21):

$$G\left(\{0\}\right) = F(\underline{v}_M) \qquad \text{and} \qquad G\big(Q(v)\big) = F(v) \quad \text{for } v \in [\underline{v}_M, \overline{v}], \tag{C6}$$

The priority fee is trivially zero for $Q_i = 0$, i.e., when $v_i \leq \underline{v}_M$.

Next, fix $v_i \in [\underline{v}_M, \overline{v}]$. Note that the atom at zero causes no problem because the integrand u^2 evaluated at zero is zero. Thus, the mass $G(\{0\})$ contributes zero to the Stieltjes integral,

and we write

$$\int_0^{Q_i} u^2\, dG(u) = 0\, G(\{0\}) + \int_{(0,Q_i]} u^2\, dG(u) = \int_{\underline{v}_M}^{v_i} Q(x)^2\, dF(x)\,. \tag{C7}$$

where we used that $Q : [\underline{v}_M, \overline{v}] \mapsto [0, Q(\overline{v})]$, defined in (18), is increasing and continuous, hence it is invertible on its image and $G(Q(v)) = F(v)$ for $v \geq \underline{v}_M$. Substitute

$$Q(x) = L\,\tilde{Q}(x) \quad \Longrightarrow \quad Q(x)^2 = L^2\,\tilde{Q}(x)^2,$$

and combine with the coefficient $2/L$ from (12) to obtain (20).

Because $\tilde{Q}$ in (19) does not depend on L, $\Phi(v_i)$ is linear in L by (20), hence increasing in L. Next, we show that for all $v_i < \overline{v}$, the priority fee tends to zero when the number of competitors M tends to infinity. For $v_i \leq \underline{v}_M$, $\Phi(v_i) = 0$ by the first part of the proposition. We therefore consider $v_i \in (\underline{v}_M, \overline{v})$ in what follows. Consider the strategic volumes (19):

$$\tilde{Q}(x) = \frac{1}{2}\Big((\overline{v} - \pi)e^{-(M-1)(1-F(x))} - \int_x^{\overline{v}} e^{-(M-1)(F(u)-F(x))}\, du\Big) \quad \text{if } x \geq \underline{v}_M,$$

and zero otherwise. By Proposition 2, $\tilde{Q}(x) \geq 0$ on $[\underline{v}_M, \overline{v}]$, and the second term in $\tilde{Q}(x)$ is nonnegative, so

$$0 \leq \tilde{Q}(x) \leq \frac{\overline{v} - \pi}{2}\, e^{-(M-1)(1-F(x))} \implies \tilde{Q}(x)^2 \leq \frac{(\overline{v} - \pi)^2}{4} e^{-2(M-1)(1-F(x))}.$$

Fix $v_i < \overline{v}$. F is nondecreasing, so for all $x \leq v_i$ we have $F(x) \leq F(v_i)$ and therefore

$$1 - F(x) \geq 1 - F(v_i) > 0.$$

Thus,

$$\tilde{Q}(x)^2 \leq \frac{(\overline{v} - \pi)^2}{4} e^{-2(M-1)(1-F(v_i))} \quad \text{for all } x \in [\underline{v}_M, v_i].$$

It follows from the definition of the equilibrium priority fee in (20), accounting for strategic volumes, that

$$\begin{aligned}\Phi(v_i) &= 2L(M-1)\int_{\underline{v}_M}^{v_i} \tilde{Q}(x)^2\, dF(x) \\ &\leq 2L(M-1)(F(v_i)-F(\underline{v}_M))\frac{(\overline{v}-\pi)^2}{4}e^{-2(M-1)(1-F(v_i))}. \end{aligned} \tag{C8}$$

Because $F(v_i) - F(\underline{v}_M) \leq 1$, we obtain

$$0 \leq \Phi(v_i) \leq \frac{L(\overline{v}-\pi)^2}{2}(M-1)e^{-2(M-1)(1-F(v_i))}.$$

Because $1 - F(v_i) > 0$ for all $v_i < \overline{v}$, we conclude that, for all $v_i < \overline{v}$,

$$\lim_{M\to\infty} \Phi(v_i) = 0.$$

□

F. Proof of Proposition 4

We first establish the closed form (23). The first-order condition (C2) characterizing the optimal trading volume is

$$0 \;=\; (v_i - \pi)\, L - 2\, Q_i - 2\,(M-1)\int_{Q_i}^{\overline{Q}} x\, dG(x).$$

Evaluate this condition at $v_i = \underline{v}_M$, and use $Q(\underline{v}_M) = 0$ from Proposition 2, to obtain

$$(\underline{v}_M - \pi)\, L \;=\; 2\,(M-1)\int_{0}^{\overline{Q}} x\, dG(x). \tag{C9}$$

By (21) and (18), the atom $G(\{0\}) = F(\underline{v}_M)$ contributes zero to the Stieltjes integral (the integrand x is zero at $x = 0$), and the change of variable

$$x = Q(v) = L\,\tilde{Q}(v)$$

on $[\underline{v}_M, \overline{v}]$ gives

$$\int_0^{\overline{Q}} x\,dG(x) \;=\; \int_{\underline{v}_M}^{\overline{v}} Q(v)\,dF(v) \;=\; L\int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(v)\,dF(v). \tag{C10}$$

Combine (C9) and (C10) to obtain

$$(\underline{v}_M - \pi)\,L \;=\; 2\,(M-1)\,L\int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(v)\,dF(v),$$

then multiply by $M/(2(M-1))$ to obtain (23). The limit $L(\overline{v}-\pi)/2$ then follows directly from (23) and Lemma 3, which gives $\lim_{M\to\infty} \underline{v}_M = \overline{v}$.

Recall that for $v_i \geq \underline{v}_M$,

$$\tilde{Q}(v_i) = \frac{1}{2}\left((\overline{v}-\pi)e^{-(M-1)(1-F(v_i))} - \int_{v_i}^{\overline{v}} e^{-(M-1)(F(u)-F(v_i))}\,du\right). \tag{C11}$$

Factor out $e^{-(M-1)(1-F(v_i))}$ to obtain

$$\tilde{Q}(v_i) = \frac{1}{2}e^{-(M-1)(1-F(v_i))}\left(\overline{v}-\pi-\int_{v_i}^{\overline{v}} e^{(M-1)(1-F(u))}\,du\right).$$

Hence, the expected total trading volume of informed traders is

$$M\int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(x)\,dF(x) = \frac{M(\overline{v}-\pi)}{2}\int_{\underline{v}_M}^{\overline{v}} e^{-(M-1)(1-F(x))}\,dF(x) \tag{C12}$$

$$-\frac{M}{2}\int_{\underline{v}_M}^{\overline{v}} e^{-(M-1)(1-F(x))}\left[\int_x^{\overline{v}} e^{(M-1)(1-F(u))}\,du\right]dF(x). \tag{C13}$$

We first evaluate (C12). Use the change of variable $z = 1 - F(x)$ to obtain

$$\begin{aligned}\frac{M(\overline{v}-\pi)}{2}\int_{\underline{v}_M}^{\overline{v}} e^{-(M-1)(1-F(x))}\,dF(x) &= \frac{M(\overline{v}-\pi)}{2}\int_0^{1-F(\underline{v}_M)} e^{-(M-1)z}\,dz \\ &= \frac{M(\overline{v}-\pi)}{2(M-1)}\left(1-e^{-(M-1)(1-F(\underline{v}_M))}\right). \qquad \text{(C14)}\end{aligned}$$

We now show that

$$\lim_{M\to\infty}(M-1)\big(1-F(\underline{v}_M)\big) = \infty.$$

Suppose instead that this sequence is bounded. Then there exists $C > 0$ such that

$$(M-1)\big(1-F(\underline{v}_M)\big) < C \qquad \text{for all } M.$$

Use the definition of $\underline{v}_M$ in (17),

$$\underline{v}_M - \pi = \int_{\underline{v}_M}^{\overline{v}} \big(e^{(M-1)(1-F(u))} - 1\big)\,du,$$

and the bound $1 - F(u) \leq 1 - F(\underline{v}_M)$, to obtain

$$0 \leq \underline{v}_M - \pi \leq \big(e^{(M-1)(1-F(\underline{v}_M))} - 1\big)(\overline{v} - \underline{v}_M) \leq \big(e^C - 1\big)(\overline{v} - \underline{v}_M).$$

By Lemma 3, $\underline{v}_M \to \overline{v}$ as $M \to \infty$. The inequality above would then imply $\underline{v}_M \to \pi$, which contradicts Lemma 3. Hence,

$$(M-1)\big(1-F(\underline{v}_M)\big) \to \infty. \qquad \text{(C15)}$$

Combine this result with (C14) to obtain

$$\lim_{M\to\infty}\frac{M(\overline{v}-\pi)}{2}\int_{\underline{v}_M}^{\overline{v}} e^{-(M-1)(1-F(x))}\,dF(x) = \frac{\overline{v}-\pi}{2}.$$

We now turn to (C13). Define

$$I_2 = \frac{M}{2} \int_{\underline{v}_M}^{\overline{v}} e^{-(M-1)(1-F(x))} \left[\int_x^{\overline{v}} e^{(M-1)(1-F(u))}\, du \right] dF(x).$$

Reverse the order of integration to obtain

$$I_2 = \frac{M}{2} \int_{\underline{v}_M}^{\overline{v}} e^{(M-1)(1-F(u))} \left[\int_{\underline{v}_M}^{u} e^{-(M-1)(1-F(x))}\, dF(x) \right] du.$$

With the change of variable $z = 1 - F(x)$,

$$\begin{aligned} \int_{\underline{v}_M}^{u} e^{-(M-1)(1-F(x))}\, dF(x) &= \int_{1-F(u)}^{1-F(\underline{v}_M)} e^{-(M-1)z}\, dz \\ &= \frac{1}{M-1} \left(e^{-(M-1)(1-F(u))} - e^{-(M-1)(1-F(\underline{v}_M))} \right). \end{aligned}$$

Substitute back to obtain

$$I_2 = \frac{M}{2(M-1)} \int_{\underline{v}_M}^{\overline{v}} \left(1 - e^{-(M-1)(1-F(\underline{v}_M))} e^{(M-1)(1-F(u))} \right) du.$$

Hence,

$$I_2 = \frac{M}{2(M-1)} \left[(\overline{v} - \underline{v}_M) - e^{-(M-1)(1-F(\underline{v}_M))} \int_{\underline{v}_M}^{\overline{v}} e^{(M-1)(1-F(u))}\, du \right].$$

Rearrange the cutoff equation (17) to obtain the exact identity

$$\int_{\underline{v}_M}^{\overline{v}} e^{(M-1)(1-F(u))}\, du \;=\; \overline{v} - \pi.$$

Because $(M-1)(1-F(\underline{v}_M)) \to \infty$ (see (C15)), the exponential term vanishes, and therefore $I_2 \to 0$ as $M \to \infty$.

Combine the limits of (C12) and (C13) to conclude that

$$\lim_{M\to\infty} M\int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(u)\,dF(u) = \frac{\overline{v}-\pi}{2}.$$

Next, we show that the aggregate trading volume approaches its limit from below; that is,

$$L\,M\int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(x)\,dF(x) \;<\; L\,\frac{\overline{v}-\pi}{2} \tag{C16}$$

for all M sufficiently large. By the standing assumption (22), there exists $\eta > 0$ such that

$$f(u) \le 1/(\overline{v}-\pi) \quad \text{for all} \quad u \in [\overline{v}-\eta, \overline{v}].$$

We establish (C16) by comparison with the uniform distribution on $[\pi, \overline{v}]$. Let F_U denote the uniform CDF on $[\pi, \overline{v}]$, i.e.,

$$F_U(u) = (u-\pi)/(\overline{v}-\pi),$$

and let $\underline{v}_M^U$ denote the corresponding cutoff defined by (17) with F replaced by F_U.

We first compute the uniform cutoff and the corresponding uniform aggregate volume. Substitute F_U in (17) and use the change of variable $s = (\overline{v}-u)/(\overline{v}-\pi)$ to obtain

$$\underline{v}_M^U - \pi \;=\; \int_{\underline{v}_M^U}^{\overline{v}} \big(e^{(M-1)(\overline{v}-u)/(\overline{v}-\pi)} - 1\big)\,du \;=\; \frac{\overline{v}-\pi}{M-1}\big(e^{(M-1)(\overline{v}-\underline{v}_M^U)/(\overline{v}-\pi)} - 1\big) - (\overline{v}-\underline{v}_M^U),$$

which simplifies to

$$e^{(M-1)(\overline{v}-\underline{v}_M^U)/(\overline{v}-\pi)} = M \quad \Longrightarrow \quad \underline{v}_M^U \;=\; \overline{v} - (\overline{v}-\pi)\,\frac{\log M}{M-1}. \tag{C17}$$

Substitute (C17) into the closed form (23) applied to F_U to obtain

$$L\,M\int_{\underline{v}_M^U}^{\overline{v}}\tilde{Q}(x)\,dF_U(x)\;=\;L\,\frac{\overline{v}-\pi}{2}\,\frac{M\big(M-1-\log M\big)}{(M-1)^2}. \tag{C18}$$

The standard inequality $\log M > 1 - 1/M$ for $M > 1$ is equivalent to

$$M(M-1-\log M) < (M-1)^2.$$

Hence, for all $M > 1$,

$$L\,M\int_{\underline{v}_M^U}^{\overline{v}}\tilde{Q}(x)\,dF_U(x)\;<\;L\,\frac{\overline{v}-\pi}{2}. \tag{C19}$$

Next, we establish tail dominance of F over F_U. Assume M is large enough that

$$(\overline{v}-\pi)\log M/(M-1) < \eta,$$

which is equivalent to $\underline{v}_M^U > \overline{v} - \eta$ by (C17). For any $u \in [\underline{v}_M^U, \overline{v}] \subset [\overline{v}-\eta, \overline{v}]$, integrate the condition $f(u) \le 1/(\overline{v}-\pi)$ in (22) to obtain

$$1-F(u)\;=\;\int_u^{\overline{v}} f(s)\,ds\;\le\;\frac{\overline{v}-u}{\overline{v}-\pi}\;=\;1-F_U(u). \tag{C20}$$

We now compare the cutoffs $\underline{v}_M$ and $\underline{v}_M^U$. For any continuous CDF G on $[\pi, \overline{v}]$ and $v \in [\pi, \overline{v}]$, define

$$\mathcal{I}(v;G)\;:=\;\int_v^{\overline{v}}\big(e^{(M-1)(1-G(u))}-1\big)\,du.$$

By (17), $\underline{v}_M$ is the unique root in $[\pi, \overline{v})$ of $v \mapsto \mathcal{I}(v;F) - (v-\pi)$. This map is strictly decreasing, because its derivative equals

$$-e^{(M-1)(1-F(v))} < 0.$$

Evaluate at $v = \underline{v}_M^U$ and use (C20) together with the monotonicity of $x \mapsto e^{(M-1)x}$ to obtain

$$\mathcal{I}(\underline{v}_M^U; F) \;\leq\; \mathcal{I}(\underline{v}_M^U; F_U) \;=\; \underline{v}_M^U - \pi,$$

where the last equality is (17) applied to F_U at its cutoff $\underline{v}_M^U$. Thus

$$\mathcal{I}(\underline{v}_M^U; F) - (\underline{v}_M^U - \pi) \leq 0,$$

and the strict monotonicity gives

$$\underline{v}_M \;\leq\; \underline{v}_M^U. \tag{C21}$$

We now conclude. Use the closed form (23), (C21), and (C19) to obtain

$$L\,M \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(x)\, dF(x) \;=\; \frac{L\,M\,(\underline{v}_M - \pi)}{2\,(M-1)} \;\leq\; \frac{L\,M\,(\underline{v}_M^U - \pi)}{2\,(M-1)} \;<\; L\,\frac{\overline{v} - \pi}{2},$$

which establishes (C16) for every M satisfying

$$(\overline{v} - \pi) \log M/(M-1) < \eta.$$

We finally establish convergence of the expected end-of-block price. Each unit of volume executed in the block moves the DEX price by $2/L$; see (A3). The initial price is normalized to zero, so the expected end-of-block price is

$$\frac{2}{L}\,M \int_{\underline{v}_M}^{\overline{v}} Q(v)\, dF(v) \;=\; 2\,M \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(v)\, dF(v) \;=\; \frac{M(\underline{v}_M - \pi)}{M-1},$$

where the last equality uses the closed form (23). By Lemma 3, $\underline{v}_M \to \overline{v}$, so the expected end-of-block price converges to $\overline{v} - \pi$ as $M \to \infty$. The expected end-of-block price equals

$2/L$ times the expected aggregate trading volume, so (C16) gives

$$M(\underline{v}_M - \pi)/(M-1) < \overline{v} - \pi$$

for all M sufficiently large; hence the price limit is also approached from below. □

G. Proof of Proposition 5

We first derive $\mathbb{V}[\Delta_M]$. Recall the liquidity supplier's stage-two prior stated in Section III. With probability 1/2, the liquidation value satisfies $V > 0$ and the informed traders' valuations $\{v_1, \ldots, v_M\}$ are i.i.d. draws from the density f on $[0, \overline{v}]$ (each trader buys); and with probability 1/2, $V < 0$ and valuations are i.i.d. draws from the symmetric density on $[-\overline{v}, 0]$ (each trader sells). Equilibrium volumes are

$$Q(v_k) = L\,\tilde{Q}(v_k)$$

when $V > 0$ by Proposition 2, and $-Q(|v_k|)$ when $V < 0$ (see Section II.E).
By construction, the conditional distribution of $Q(v_k)$ on $\{V < 0\}$ equals the negative of its conditional distribution on $\{V > 0\}$, so

$$\mathbb{E}[Q(v_k)] \;=\; \tfrac{1}{2}\,\mathbb{E}[Q(v_k) \mid V > 0] + \tfrac{1}{2}\,\mathbb{E}[Q(v_k) \mid V < 0] \;=\; 0,$$

hence

$$\mathbb{E}[\Delta_M] = 0 \qquad \text{and} \qquad \mathbb{V}[\Delta_M] = \mathbb{E}[\Delta_M^2].$$

Conditional on the sign of V, the valuations v_k are i.i.d., so the volumes $Q(v_k)$ are conditionally i.i.d. as well. Expand the square to obtain

$$\mathbb{E}[\Delta_M^2] \;=\; M\,\mathbb{E}[Q(v_k)^2] \;+\; M(M-1)\,\mathbb{E}[Q(v_k)\,Q(v_l)] \qquad (k \neq l). \tag{C22}$$

For the diagonal term, use (Proposition 2):

$$Q(v_k) = L\,\tilde{Q}(v_k) \quad \text{on} \quad \{v_k \geq \underline{v}_M\} \qquad \text{and} \qquad Q(v_k) = 0 \quad \text{on} \quad \{v_k < \underline{v}_M\},$$

and the symmetry of the liquidity supplier's prior across signs of V, to obtain

$$\mathbb{E}[Q(v_k)^2] \;=\; \mathbb{E}[Q(v_k)^2 \mid V > 0] \;=\; L^2 \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(u)^2\, dF(u). \tag{C23}$$

For the cross term, with $k \neq l$, conditional independence gives, on each side of the prior,

$$\mathbb{E}[Q(v_k)\,Q(v_l) \mid V > 0] \;=\; \big(\mathbb{E}[Q(v_k) \mid V > 0]\big)^2 \;=\; \Big(L \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(u)\, dF(u)\Big)^2,$$

and by symmetry

$$\mathbb{E}[Q(v_k)\,Q(v_l) \mid V < 0] = (-1)^2 \big(\mathbb{E}[Q(v_k) \mid V > 0]\big)^2$$

equals the same value. Use the closed form (23),

$$\int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(u)\, dF(u) = (\underline{v}_M - \pi)/(2(M-1))$$

to obtain

$$\mathbb{E}[Q(v_k)\,Q(v_l)] \;=\; L^2\, \frac{(\underline{v}_M - \pi)^2}{4\,(M-1)^2}. \tag{C24}$$

Substitute (C23)–(C24) into (C22) to obtain

$$\mathbb{V}[\Delta_M] \;=\; L^2\, M \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(u)^2\, dF(u) \;+\; L^2\, \frac{M\,(\underline{v}_M - \pi)^2}{4\,(M-1)} \;=\; L^2\, M\, S_M, \tag{C25}$$

where the last equality uses the definition (30). Combined with $\mathbb{E}[\Delta_M] = 0$, this gives the rightmost equality in (29).

Next, we derive the zero-profit closed form. Multiply the zero-profit condition (31) by

$(L^\star + \theta)/L^\star$ (with $L^\star > 0$, addressed below) to obtain

$$\pi\, N \;=\; (L^\star + \theta)\, M\, S_M,$$

which rearranges to the closed form (32),

$$L^\star(M) = \pi\, N/(M\, S_M) - \theta.$$

The supply increases in N, because S_M is independent of N.

We next establish the viability condition. Non-negativity of (32) requires

$$\pi\, N/(M\, S_M) \geq \theta,$$

which rearranges to condition (34).

We now derive the asymptotic limit. By (30), we have

$$M\, S_M \;=\; M \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(u)^2\, dF(u) \;+\; \frac{M\,(\underline{v}_M - \pi)^2}{4\,(M-1)}. \tag{C26}$$

The second term on the RHS of (C26) converges to

$$\frac{M\,(\underline{v}_M - \pi)^2}{4\,(M-1)} \to (\overline{v} - \pi)^2/4 \tag{C27}$$

because, by Lemma 3, $\underline{v}_M \to \overline{v}$ and $M/(M-1) \to 1$. For the first term on the RHS of (C26), we need to show

$$\lim_{M\to\infty} M \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(u)^2\, dF(u) = (\overline{v} - \pi)^2/8,$$

which combined with the limit (C27) gives

$$\lim_{M\to\infty} M\, S_M = 3\,(\overline{v}-\pi)^2/8,$$

hence by (32),

$$\lim_{M\to\infty} L^\star(M) \;=\; \frac{8\,\pi\,N}{3\,(\overline{v}-\pi)^2} - \theta,$$

which is (33). To distinguish the two contributions to $M\,S_M$ in (C26), denote

$$\widehat{S}_M \;:=\; \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(u)^2\, dF(u) \qquad \text{so that} \qquad S_M \;=\; \widehat{S}_M \;+\; \frac{(\underline{v}_M-\pi)^2}{4\,(M-1)}. \tag{C28}$$

It remains to show that

$$\lim_{M\to\infty} M\,\widehat{S}_M = (\overline{v}-\pi)^2/8.$$

Use the factored form (C11) of $\tilde{Q}$ (Proposition 4) to write

$$\tilde{Q}(x) = \frac{1}{2}\, e^{-(M-1)(1-F(x))}\,\left((\overline{v}-\pi) - B(x)\right), \qquad B(x) \equiv \int_x^{\overline{v}} e^{(M-1)(1-F(u))}\, du,$$

so that

$$M\,\widehat{S}_M = \frac{M}{4}\int_{\underline{v}_M}^{\overline{v}} e^{-2(M-1)(1-F(x))}\,\left((\overline{v}-\pi) - B(x)\right)^2 dF(x) = T_1 + T_2 + T_3, \tag{C29}$$

where

$$\begin{aligned}
T_1 &= \frac{M(\overline{v}-\pi)^2}{4}\int_{\underline{v}_M}^{\overline{v}} e^{-2(M-1)(1-F(x))}\, dF(x),\\
T_2 &= -\frac{M(\overline{v}-\pi)}{2}\int_{\underline{v}_M}^{\overline{v}} e^{-2(M-1)(1-F(x))}\, B(x)\, dF(x),\\
T_3 &= \frac{M}{4}\int_{\underline{v}_M}^{\overline{v}} e^{-2(M-1)(1-F(x))}\, B(x)^2\, dF(x).
\end{aligned}$$

For T_1, the change of variable $z = 1 - F(x)$ gives

$$T_1 = \frac{M(\overline{v} - \pi)^2}{4} \int_0^{1-F(\underline{v}_M)} e^{-2(M-1)z}\, dz = \frac{M(\overline{v} - \pi)^2}{8(M-1)} \left(1 - e^{-2(M-1)(1-F(\underline{v}_M))}\right). \tag{C30}$$

Because, as established in the proof of Proposition 4,

$$(M-1)(1 - F(\underline{v}_M)) \to \infty,$$

the exponential term in (C30) goes to zero and $M/(M-1) \to 1$, so

$$T_1 \to \frac{(\overline{v} - \pi)^2}{8}. \tag{C31}$$

For T_2, reverse the order of integration between x and u to obtain

$$\int_{\underline{v}_M}^{\overline{v}} e^{-2(M-1)(1-F(x))} B(x)\, dF(x) = \int_{\underline{v}_M}^{\overline{v}} e^{(M-1)(1-F(u))} \left[\int_{\underline{v}_M}^{u} e^{-2(M-1)(1-F(x))}\, dF(x)\right] du.$$

The inner integral evaluates, using the change of variables $z = 1 - F(x)$, to

$$\int_{\underline{v}_M}^{u} e^{-2(M-1)(1-F(x))}\, dF(x) = \frac{1}{2(M-1)} \left(e^{-2(M-1)(1-F(u))} - e^{-2(M-1)(1-F(\underline{v}_M))}\right).$$

Substitute back and use the cutoff definition (17)

$$\int_{\underline{v}_M}^{\overline{v}} e^{(M-1)(1-F(u))}\, du = \overline{v} - \pi,$$

to obtain

$$T_2 = -\frac{M(\overline{v} - \pi)}{4(M-1)} \left[\int_{\underline{v}_M}^{\overline{v}} e^{-(M-1)(1-F(u))}\, du \;-\; (\overline{v} - \pi)\, e^{-2(M-1)(1-F(\underline{v}_M))}\right].$$

The first term in brackets is bounded by $\overline{v} - \underline{v}_M \to 0$; the second goes to zero because

$(M-1)(1-F(\underline{v}_M)) \to \infty$. Hence

$$T_2 \to 0. \tag{C32}$$

For T_3, because B is decreasing in x, we have

$$B(x) \leq B(\underline{v}_M) = \overline{v} - \pi \quad \text{for all} \quad x \in [\underline{v}_M, \overline{v}],$$

so

$$0 \leq T_3 \leq \frac{(\overline{v}-\pi)}{4} M \int_{\underline{v}_M}^{\overline{v}} e^{-2(M-1)(1-F(x))} B(x)\, dF(x) \to 0, \tag{C33}$$

where the last limit follows from the computation for T_2.

Combine the three limits (C30), (C32), and (C33) to obtain

$$\lim_{M\to\infty} M\, \widehat{S}_M = \frac{(\overline{v}-\pi)^2}{8},$$

which together with the limit

$$\lim_{M\to\infty} M\, (\underline{v}_M - \pi)^2/(4(M-1)) = (\overline{v}-\pi)^2/4$$

established above gives

$$\lim_{M\to\infty} M\, S_M = 3\,(\overline{v}-\pi)^2/8,$$

and hence establishes the result (33).

Finally, we establish convergence from above by showing that

$$M\, S_M \;<\; \frac{3\,(\overline{v}-\pi)^2}{8} \tag{C34}$$

for all M sufficiently large, which by (32) is equivalent to

$$L^\star(M) > \lim_{M\to\infty} L^\star(M).$$

By (C26), it suffices to prove the two strict inequalities

$$M\,\widehat{S}_M \;<\; \frac{(\overline{v}-\pi)^2}{8}, \tag{C35}$$

$$\frac{M\,(\underline{v}_M-\pi)^2}{4\,(M-1)} \;<\; \frac{(\overline{v}-\pi)^2}{4}. \tag{C36}$$

We first prove (C35). Apply the change of variable $z = 1 - F(x)$ to the expression for $M\,\widehat{S}_M$ in (C29) to obtain

$$M\,\widehat{S}_M \;=\; \frac{M}{4}\int_0^{1-F(\underline{v}_M)} e^{-2(M-1)z}\big((\overline{v}-\pi)-B(x)\big)^2\,dz, \tag{C37}$$

where we used $B(\overline{v}) = 0$ and, by (17), $B(\underline{v}_M) = \overline{v} - \pi$.

By the standing assumption (22), there exists $\eta > 0$ such that

$$f \le 1/(\overline{v}-\pi) \quad \text{on} \quad [\overline{v}-\eta, \overline{v}].$$

Use $\underline{v}_M \to \overline{v}$ (Proposition 4) to obtain that for M large enough, we have

$$\underline{v}_M \ge \overline{v}-\eta \quad \Longrightarrow \quad 1/f(u) \ge \overline{v}-\pi \quad \text{for all} \quad u \in [\underline{v}_M, \overline{v}].$$

We first derive a lower bound on B in (C37). For $z \in [0, 1-F(\underline{v}_M)]$ and $x = F^{-1}(1-z)$, the change of variable $s = 1 - F(u)$ in

$$B(x) = \int_x^{\overline{v}} e^{(M-1)(1-F(u))}\,du$$

gives

$$B(x) \;=\; \int_0^z \frac{e^{(M-1)s}}{f(u)}\,ds \tag{C38}$$

$$\geq\; (\overline{v}-\pi)\int_0^z e^{(M-1)s}\,ds \;=\; \frac{\overline{v}-\pi}{M-1}\big(e^{(M-1)z}-1\big), \tag{C39}$$

where the inequality uses $1/f(u) \geq \overline{v}-\pi$ on $[\underline{v}_M, \overline{v}]$.

Evaluate the lower bound in (C38) at $z = 1-F(\underline{v}_M)$ (so $x = \underline{v}_M$) and use $B(\underline{v}_M) = \overline{v}-\pi$ to obtain

$$\overline{v}-\pi \geq (\overline{v}-\pi)\big(e^{(M-1)(1-F(\underline{v}_M))}-1\big)/(M-1),$$

which rearranges to

$$1-F(\underline{v}_M) \;\leq\; \frac{\log M}{M-1}. \tag{C40}$$

Rearrange the lower bound (C38) to obtain

$$(\overline{v}-\pi)-B(x) \;\leq\; (\overline{v}-\pi)-\frac{\overline{v}-\pi}{M-1}\big(e^{(M-1)z}-1\big) \;=\; \frac{(\overline{v}-\pi)\big(M-e^{(M-1)z}\big)}{M-1}.$$

The left-hand side is nonnegative because B is decreasing in x so

$$B(x) \leq B(\underline{v}_M) = \overline{v}-\pi.$$

The right-hand side is nonnegative by (C40). Therefore, squaring preserves the inequality and we write

$$\big((\overline{v}-\pi)-B(x)\big)^2 \;\leq\; \frac{(\overline{v}-\pi)^2}{(M-1)^2}\big(M-e^{(M-1)z}\big)^2 \quad \text{on} \quad [0, 1-F(\underline{v}_M)]. \tag{C41}$$

Insert (C41) into (C37) and extend the integration range from $[0, 1-F(\underline{v}_M)]$ to $[0, \log M/(M-$

1)] to obtain

$$M\,\widehat{S}_M \;\le\; \frac{M(\overline{v}-\pi)^2}{4(M-1)^2}\int_0^{\log M/(M-1)} e^{-2(M-1)z}\big(M-e^{(M-1)z}\big)^2\,dz.$$

Note that the extension of the integration range is valid by (C40), because the integrand is nonnegative. The bound is solved to

$$M\,\widehat{S}_M \;\le\; \frac{(\overline{v}-\pi)^2}{8}\left[\frac{M(M-3)}{(M-1)^2}+\frac{2\,M\,\log M}{(M-1)^3}\right]. \tag{C42}$$

It remains to verify the strict inequality. Define

$$\nu := M-1 \ge 1 \qquad \text{and} \qquad g(\nu) := \nu(\nu+2)-2(\nu+1)\log(\nu+1).$$

The bracketed term in (C42) is strictly less than 1 if and only if $g(\nu)>0$. We have

$$g'(\nu)=2\nu-2\log(\nu+1), \quad g'(0)=0, \quad \text{and} \quad g''(\nu)=2\nu/(\nu+1)\ge 0.$$

Hence g is non-decreasing on $[0,\infty)$. Use

$$g(1)=3-4\log 2>0,$$

and monotonicity to obtain

$$g(\nu)\ge g(1)>0 \quad \text{for all} \quad \nu\ge 1.$$

Combine with (C42), to establish (C35) for every M with $\underline{v}_M \ge \overline{v}-\eta$.

We now prove (C36). Under the standing assumption $\underline{v}_M \ge \overline{v}-\eta$, use (C40) together with

the cutoff identity (17) to bound $\overline{v} - \underline{v}_M$ from below as follows. First,

$$F(u) \geq F(\underline{v}_M) \quad \text{for} \quad u \in [\underline{v}_M, \overline{v}].$$

Therefore, by (C40),

$$(M-1)(1-F(u)) \leq (M-1)(1-F(\underline{v}_M)) \leq \log M \quad \Longrightarrow \quad e^{(M-1)(1-F(u))} - 1 \leq M - 1.$$

Substitute into (17) to obtain

$$\underline{v}_M - \pi \;=\; \int_{\underline{v}_M}^{\overline{v}} \left(e^{(M-1)(1-F(u))} - 1\right) du \;\leq\; (M-1)\,(\overline{v} - \underline{v}_M),$$

which rearranges to

$$\overline{v} - \underline{v}_M \;\geq\; \frac{\underline{v}_M - \pi}{M-1}. \tag{C43}$$

Define

$$\gamma := (\underline{v}_M - \pi)/(\overline{v} - \pi) \in (0, 1].$$

The inequality (C43) divided by $\overline{v} - \pi$ gives

$$1 - \gamma \geq \gamma/(M-1) \quad \Longrightarrow \quad (M-1)(1-\gamma) \geq \gamma \quad \Longrightarrow \quad \gamma \leq (M-1)/M < 1.$$

Therefore,

$$\begin{aligned} \frac{M\,(\underline{v}_M - \pi)^2}{4\,(M-1)} \;&=\; \frac{M\,\gamma^2\,(\overline{v} - \pi)^2}{4\,(M-1)} \\ &\leq\; \frac{M\,(M-1)^2/M^2\,(\overline{v} - \pi)^2}{4\,(M-1)} \;=\; \frac{(M-1)\,(\overline{v} - \pi)^2}{4\,M} \;<\; \frac{(\overline{v} - \pi)^2}{4}, \end{aligned}$$

which proves (C36). Combine (C35) and (C36) using (C26) to obtain (C34) for every M with $\underline{v}_M \geq \overline{v} - \eta$, hence $L^\star(M)$ is approached from above as $M \to \infty$. □

H. Proof of Proposition 6

Use the expected wealth of trader i, conditional on observing a valuation v_i, in (5),

$$\mathbb{E}_i\big[W_i\big] = -\Phi_i + Q_i\Big(v_i - \pi - \frac{Q_i}{L}\Big) - C - \frac{2\,Q_i}{L}\sum_{j=0}^{M-1}\mathbb{E}_i\Big[\mathbb{1}_{\{\Phi_{(j)}<\Phi_i<\Phi_{(j+1)}\}}\,\Delta_{(j+1:M-1)}\Big]\,,$$

together with the equilibrium priority fees (Proposition 3), trading volumes (Proposition 2), and liquidity (Proposition 5), to obtain for each $i \in \{1,\dots,M\}$:

$$\begin{aligned}\mathbb{E}\big[W_i \mid v_i\big] = &-\frac{2}{L^\star}\,(M-1)\left(\int_0^{Q^\star(v_i)} x^2\,dG(x) + Q^\star(v_i)\int_{Q^\star(v_i)}^{\overline{Q}} x\,dG(x)\right)\\ &+ Q^\star(v_i)\left(v_i - \pi - \frac{Q^\star(v_i)}{L^\star}\right) - C.\end{aligned}$$

Use that

$$Q^\star(v_i) = 0 \quad \text{for} \quad v_i < \underline{v}_M \qquad \text{and} \qquad Q^\star(v_i) = L^\star\tilde{Q}(v_i) \quad \text{for} \quad v_i \geq \underline{v}_M$$

to write the expectation as

$$\begin{aligned}\mathbb{E}\big[W_i \mid v_i\big] = L^\star\Bigg[&\tilde{Q}(v_i)\big(v_i - \pi - \tilde{Q}(v_i)\big)\\ &- 2\,(M-1)\left(\int_0^{v_i}\tilde{Q}(u)^2\,dF(u) + \tilde{Q}(v_i)\int_{v_i}^{\overline{v}}\tilde{Q}(u)\,dF(u)\right)\Bigg] - C.\end{aligned}$$

At stage one, valuations have not yet been realized and traders are ex ante symmetric. If $v_i < \underline{v}_M$, then $Q^\star(v_i) = 0$ and $\Phi_i = 0$, so the payoff is $-C$. Combine with the case $v_i \geq \underline{v}_M$ to obtain the expected wealth

$$\begin{aligned}\mathbb{E}[W_i] = &-C + L^\star\int_{\underline{v}_M}^{\overline{v}}\tilde{Q}(v)\big(v - \pi - \tilde{Q}(v)\big)\,dF(v)\\ &- 2\,L^\star(M-1)\int_{\underline{v}_M}^{\overline{v}}\left[\int_{\underline{v}_M}^{v}\tilde{Q}(u)^2\,dF(u) + \tilde{Q}(v)\int_{v}^{\overline{v}}\tilde{Q}(u)\,dF(u)\right]dF(v).\end{aligned}$$

We now show that $\mathbb{E}[W_i]$ converges to a finite negative limit as $M \to \infty$. First, by Proposition 5, $L^\star$ converges to a finite constant as $M \to \infty$.

Consider

$$(M-1)\int_{\underline{v}_M}^{\overline{v}}\int_{\underline{v}_M}^{v}\tilde{Q}(u)^2\,dF(u)\,dF(v).$$

Interchange the order of integration to obtain

$$\begin{aligned}\int_{\underline{v}_M}^{\overline{v}}\int_{\underline{v}_M}^{v}\tilde{Q}(u)^2\,dF(u)\,dF(v) &= \int_{\underline{v}_M}^{\overline{v}}\int_{u}^{\overline{v}}\tilde{Q}(u)^2\,dF(v)\,dF(u)\\ &= \int_{\underline{v}_M}^{\overline{v}}\tilde{Q}(u)^2\big(1-F(u)\big)\,dF(u).\end{aligned}$$

Because $1-F(u)\leq 1$, the limit derived in the proof of Proposition 5 implies

$$0 \leq \limsup_{M\to\infty}(M-1)\int_{\underline{v}_M}^{\overline{v}}\int_{\underline{v}_M}^{v}\tilde{Q}(u)^2\,dF(u)\,dF(v) \leq \frac{(\overline{v}-\pi)^2}{8}.$$

Next, consider

$$(M-1)\int_{\underline{v}_M}^{\overline{v}}\int_{v}^{\overline{v}}\tilde{Q}(v)\tilde{Q}(u)\,dF(u)\,dF(v).$$

Use the bound $\tilde{Q}(u)\leq\tilde{Q}(\overline{v})=\frac{\overline{v}-\pi}{2}$ to obtain

$$\begin{aligned}\int_{\underline{v}_M}^{\overline{v}}\int_{v}^{\overline{v}}\tilde{Q}(v)\tilde{Q}(u)\,dF(u)\,dF(v) &\leq \frac{\overline{v}-\pi}{2}\int_{\underline{v}_M}^{\overline{v}}\tilde{Q}(v)\,(1-F(v))\,dF(v)\\ &\leq \frac{\overline{v}-\pi}{2}\int_{\underline{v}_M}^{\overline{v}}\tilde{Q}(v)\,dF(v).\end{aligned}$$

By the limit derived in Proposition 4,

$$0 \leq \limsup_{M\to\infty}(M-1)\int_{\underline{v}_M}^{\overline{v}}\int_{v}^{\overline{v}}\tilde{Q}(v)\tilde{Q}(u)\,dF(u)\,dF(v) \leq \frac{(\overline{v}-\pi)^2}{4}.$$

Collect terms to rearrange the indifference condition $\mathbb{E}[W_i] = 0$ to

$$\begin{aligned} C =& L^\star \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(v)\big(v - \pi - \tilde{Q}(v)\big)\, dF(v) \\ & - 2\, L^\star (M-1) \int_{\underline{v}_M}^{\overline{v}} (1 - F(u)) \tilde{Q}(u)^2\, dF(u) \\ & - 2\, L^\star (M-1) \int_{\underline{v}_M}^{\overline{v}} \int_{v}^{\overline{v}} \tilde{Q}(v)\tilde{Q}(u)\, dF(u)\, dF(v). \end{aligned}$$

The last two terms on the RHS of the equation above are nonnegative because $\tilde{Q} \geq 0$, and the first integral term converges to zero because its integrand is bounded uniformly in M and $F(\underline{v}_M) \to 1$. Use that $L^\star$ is nonnegative and bounded, and $C > 0$, to obtain

$$\limsup_{M \to \infty} \mathbb{E}[W_i] < 0.$$

Therefore, if $\mathbb{E}[W_i] > 0$ for $M = 2$, there exists a finite equilibrium number of informed traders, defined as the largest integer M such that $\mathbb{E}[W_i] \geq 0$.

Next, we simplify the expression of $H(M)$. Substitute $Q = L^\star \tilde{Q}$ into the volume FOC (16) and multiply by $\tilde{Q}(v)$ to obtain the identity

$$\tilde{Q}(v)\big(v - \pi - \tilde{Q}(v)\big) \;=\; \tilde{Q}(v)^2 \;+\; 2(M-1)\, \tilde{Q}(v) \int_{v}^{\overline{v}} \tilde{Q}(u)\, dF(u), \qquad v \in [\underline{v}_M, \overline{v}].$$

Substitute into the first integral of the indifference condition above. The $\tilde{Q}(v)\tilde{Q}(u)$ cross term cancels exactly with the third integral, leaving

$$C \;=\; L^\star \int_{\underline{v}_M}^{\overline{v}} \tilde{Q}(v)^2 \big[1 - 2(M-1)(1 - F(v))\big]\, dF(v) \;=\; H(M),$$

which is the form in (36).

We now establish the comparative statics. Decreasing C weakly enlarges the set $\{M \geq 2 : C \leq H(M)\}$, so $M^\star$ is weakly decreasing in C. By Proposition 5, $L^\star(M)$ is increasing in N

(with π fixed), while the bracketed integrand in (36) does not depend on N. At any M with $H(M) \geq C \geq 0$ and $L^\star(M) > 0$, the integral over that bracket is nonnegative, so raising N weakly raises $H(M)$ and thus weakly raises $M^\star$. □

I. Proof of Corollary 1

Throughout, $F(\cdot\,;T)$ denotes the distribution of private valuations with support $[0, \overline{v}(T)]$ and $N(T)$ the mass of noise-trading flow.

We first show that the participation cutoff $\underline{v}_M(T)$ is strictly increasing in T. Define

$$G(v;T) \;:=\; v - \pi - \int_v^{\overline{v}(T)} \Big(e^{(1-F(u;T))(M-1)} - 1\Big)\, du,$$

so that the cutoff $\underline{v}_M(T)$ satisfies $G(\underline{v}_M(T);T) = 0$ by (17). Differentiate in v to obtain,

$$\partial_v G(v;T) \;=\; 1 + \Big(e^{(1-F(v;T))(M-1)} - 1\Big) \;=\; e^{(1-F(v;T))(M-1)} \;>\; 0.$$

For $T' > T$, the FOSD ordering in Assumption 1(i) implies

$$F(u;T') < F(u;T) \quad \text{on} \quad [0, \overline{v}(T)],$$

so

$$e^{(1-F(u;T'))(M-1)} - 1 \;>\; e^{(1-F(u;T))(M-1)} - 1 \qquad \text{for } u \in (0, \overline{v}(T)].$$

Moreover, on $(\overline{v}(T), \overline{v}(T')]$ one has $F(u;T') < 1$, hence

$$e^{(1-F(u;T'))(M-1)} - 1 > 0.$$

Therefore

$$\int_v^{\overline{v}(T')} \Big(e^{(1-F(u;T'))(M-1)} - 1\Big)\, du \;>\; \int_v^{\overline{v}(T)} \Big(e^{(1-F(u;T))(M-1)} - 1\Big)\, du,$$

so

$$G(v;T') < G(v;T) \quad \text{for all} \quad v \in [0, \overline{v}(T)].$$

Use $\partial_v G > 0$ to obtain that the unique solution of $G(\cdot;T') = 0$ satisfies $\underline{v}_M(T') > \underline{v}_M(T)$.

We now show that the end-of-block price is strictly increasing in T in the regime with many informed traders. From Proposition 4, as $M \to \infty$ the end-of-block price converges to $\overline{v} - \pi$. With $\overline{v} = \overline{v}(T)$, the limit equals $\overline{v}(T) - \pi$, which is strictly increasing in T by Assumption 1(i).

Next, we show that the equilibrium liquidity $L^\infty(T)$ is decreasing in T. From (33) in Proposition 5,

$$L^\infty(T) \;=\; \frac{8\,\pi\,N(T)}{3\,(\overline{v}(T) - \pi)^2} - \theta.$$

By Assumption 1(i), $\overline{v}(T) - \pi$ is strictly increasing in T, so the denominator $3\,(\overline{v}(T) - \pi)^2$ is strictly increasing; by Assumption 1(ii), the numerator $8\,\pi\,N(T)$ is decreasing in T. Hence $L^\infty(T)$ is strictly decreasing in T.

Finally, we show that markets shut down for sufficiently large T. $N(T) \geq 0$ is decreasing, so

$$8\,\pi\,N(T) \leq 8\,\pi\,N(0)$$

is bounded. By Assumption 1(i), we have

$$\overline{v}(T) - \pi \to \infty \quad \Longrightarrow \quad 3\,(\overline{v}(T) - \pi)^2 \to \infty,$$

so the first term in $L^\infty(T)$ tends to zero, and $L^\infty(T) \to -\theta < 0$. Therefore, there exists $\overline{T} < \infty$ such that $L^\infty(T) \leq 0$ for all $T \geq \overline{T}$, i.e., the market shuts down. □

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